



Proposal Cover Sheet

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Software Requirements Specification for SentinaAI

Version 2.0

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of SentinaAI, an AI-driven, edge-cloud enabled smart management platform designed for large-scale convention centers and exhibition venues.

The document specifies the functional, non-functional, security, and data requirements of the system. It is a primary reference for system design, implementation, and validation, so that all stakeholders share an understanding of the system's objectives and expected performance.

Additionally, this will be the baseline for subsequent design documentation and will guide system development, integration, and testing phases to maintain traceability between requirements and system behavior.

1.2 Document Conventions

This document follows specific conventions for consistency throughout the document.

1.2.1 Formatting and Style

- Key system terms (e.g., *shall*, *should*, *may*) follow standardized requirement notation conventions as defined below.
- All requirement statements are uniquely numbered and formatted for traceability.
- Priorities for higher-level requirements are **not automatically inherited** by detailed sub-requirements; each requirement statement has its own assigned priority.

1.2.2 Requirement Convention

The following keywords are used to define the level of obligation for each requirement:

Keyword	Meaning
SHALL	Mandatory requirement that must be implemented in the current development phase.
SHOULD	Recommended requirements to be implemented where possible or in future iterations.
MAY	Optional requirement that can be included for enhancement or scalability.
TBD	To Be Determined, information not yet finalized but will be defined in later versions.

Table 1 - Requirement Keywords

Requirements are categorized as follows:

Prefix	Category
FR-XX	Functional Requirements
NFR-XX	Non-Functional Requirements
OC-XX	Other Constraints
UI-XX	User Interface
HI-XX	Hardware Interface
SI-XX	Software Interface
CI-XX	Communication Interface

Table 2- Requirements Prefix

1.3 Intended Audience and Reading Suggestions

This document is intended for a range of readers involved in the design, development, operation, and evaluation of the system. It provides a single reference to so that every team member and stakeholder has a consistent understanding of the system's functionality, scope, and expectations.

1.3.1 Intended Readers

Stakeholder	Purpose for Reading
Facility Operations Manager	Understands operational features such as HVAC, lighting, and occupancy control.
Security Operations Center (SOC) Analyst	Reviews system requirements for anomaly detection, alerts, and security analytics.
Event Administrator	Reviews capabilities related to event management, visitor zones, and analytics dashboards.
Exhibitor	Understands system data outputs such as booth engagement, heatmaps, and visitor interaction analytics.
Visitor	Uses this document only for awareness of supported user interactions, AI assistant functions, and navigation tools.
Maintenance Technician	Refers to technical requirements related to IoT device connectivity, diagnostics, and performance reliability.
Executive Management	Reviews the business objectives, high-level system scope, and expected outcomes to evaluate project ROI and feasibility.
Academic Supervisor	Consults for AI model validation, dataset specifications, and data governance processes.
Project Manager	Uses the SRS to track deliverables, scope, and requirement coverage across project phases.
Software Engineers (Backend / Frontend)	Implements all functional requirements, APIs, and interfaces as defined in the technical sections.

AI / Data Engineers	Develops and maintains ML components following data and model-related requirements.
Cloud Architect / DevOps Engineer	References infrastructure, performance, and operational requirements for deployment and cloud-edge coordination.
System Administrator	Uses for system access control, configuration management, and uptime requirements.
Quality Assurance (QA) Engineer	Uses to derive test cases and validation metrics aligned with requirement identifiers (FR, NFR, IR, SR).
UI/UX Designer	References user interaction requirements and visualization expectations for dashboards and portals.

Table 3- Stakeholder Table

1.3.2 Reading Suggestions

Technical readers will find [Sections 3](#) and [Section 4](#) most useful, as these cover the detailed system features, functional requirements, and interface specifications. Project managers and other stakeholders should focus on [Sections 1](#) and [Section 2](#), which includes the overall scope, priorities, and high-level understanding of the system. Operational users, including analysts, managers, and administrators, can refer to the feature overviews in [Section 3](#) to understand how SentinaAI supports their everyday work.

1.4 Project Scope

SentinaAI is an AI-driven data and automation platform for large-scale convention centers and exhibition venues. Its primary objective is to unify building systems, security operations, and visitor management into a single intelligent network operating between the edge and the cloud. The system provides real-time visibility and automation across domains such as security, energy management, crowd analytics, and sustainability performance for faster, data-driven decision-making and improved operational efficiency.

1.4.1 In Scope

The project will include the following key components:

1. **IoT Device Integration:** Connectivity and management of building systems and IoT devices such as cameras, HVAC controllers, sensors, and lighting systems.
2. **Edge Node and Local Intelligence:** Local inference and automation logic for low-latency decision-making.
3. **Cloud Coordination and Data Analytics:** Cloud-based analytics, visualization, and cross-node data aggregation.
4. **Security and Threat Detection Framework:** AI-driven anomaly detection for cyber and operational threats.
5. **Operational Dashboards and Visualization Tools:** Unified dashboards for operations control and monitoring.
6. **Digital Twin Environment:** Real-time 3D visualization of energy flow, occupancy, and system performance.
7. **AI Assistant and Interaction Layer:** Natural language-based interaction for simplified system control.
8. **Energy Optimization Module:** Predictive analytics for efficiency and sustainability scoring.
9. **Visitor and Exhibitor Interfaces:** Role-based portals for event insights and live updates.
10. **Behavioral and Operational Analytics:** Real-time visitor and exhibitor behavior analytics.
11. **Simulation and Testing Environment:** Controlled test environment for validating performance and reliability.

1.4.2 Out of Scope

The following items are excluded from this project phase:

1. **Integration with third-party Building Management Systems (BMS).**
2. **Multi-site or cross-venue deployment** beyond a single convention center.
3. **Custom hardware or firmware development** (existing IoT devices will be used).
4. **External data-sharing or monetization APIs.**
5. **Immersive AR/VR visualization or mobile app interfaces.**
6. **Inter-venue synchronization** (multi-tenancy within one venue is supported, but cross-venue linking is not).

1.5 References

Title	Author	Version	Date	Source
Feasibility and Planning Report	Team Cyber6	2.0	31/10/2025	Please request a copy via email: rz962@uowmail.edu.au

2 Overall Description

2.1 Product Perspective

SentinaAI is a new, unified platform developed to address the challenges of managing large, dynamic convention environments. It integrates IoT devices, artificial intelligence, and edge-cloud computing to provide real-time situational awareness, automation, and analytics.

Although SentinaAI is as a standalone solution, it has been designed to be modular and scalable. This allows it to integrate with future enterprise systems or broader smart-infrastructure ecosystems as organizational needs evolve.

2.1.1 System Context and Boundaries

Within the system boundary, SentinaAI includes:

- **Edge Processing Layer:** Performs real-time telemetry ingestion, local inference, anomaly detection, and device-level automation.
- **Cloud Coordination Layer:** Handles long-term analytics, AI model management, event intelligence, reporting, and dashboard orchestration.
- **User Interaction Layer:** Provides web-based dashboards, management consoles, and visitor-facing interfaces for information access and decision support.
- **Digital Twin Environment:** Maintains a virtual representation of venue zones, and events for simulation and situational awareness.

Elements outside the system boundary include all physical IoT devices and all human users.

2.1.2 External Interfaces

The platform will interface with:

- **IoT Devices:** CCTV cameras, HVAC systems, lighting units, occupancy sensors, environmental monitors, and similar devices that continuously send real-time data into the platform.
- **Human Users (External Stakeholders):** stakeholders such as facility managers, SOC analysts, event administrators, exhibitors, and visitors who access the system through web-based dashboards and portal interfaces.

2.2 Product Features

At a high level, SentinaAI offers the following key features:

1. IoT Device Integration

- Connects with CCTV cameras, HVAC controllers, lighting systems, occupancy sensors, and environmental monitors.
- Unified data collection and control through secure IoT communication protocols (e.g., MQTT, BACnet, Modbus).

2. Edge Intelligence and Local Automation

- Real-time inference and anomaly detection at the edge to minimize latency.
- Local automation rules (e.g., adaptive lighting or HVAC adjustments) based on occupancy and environmental conditions.

3. Cloud Coordination and Data Analytics

- Aggregates and processes data from multiple sources using a cloud backend.
- Centralized analytics and AI model retraining for performance optimization.
- Allows multi-tenancy and separate management of multiple events or operational zones within a single venue while maintaining data and access isolation.
- Provides administrators with configurable dashboards and trend insights.

4. Security and Threat Detection Framework

- Continuously monitors device and network activity to detect anomalies and cyber threats.
- Uses AI anomaly detection to identify DDoS attacks, unauthorized access, or abnormal system behavior.
- Real-time alerts, event logging, and SOC dashboard integration.

5. Digital Twin Visualization

- Maintains a real-time virtual model of the convention venue to visualize energy flow, occupancy, and system performance.
- Predictive analysis and operational simulation for testing and optimization.

6. AI Assistant and Interaction Layer

- Integrates a natural-language interface for staff and visitors to access information or event status updates.
- Supports context-aware interactions for both operational and visitor-related queries.

7. Sustainability and Energy Optimization

- Tracks and predicts energy consumption to support sustainable operations.
- Recommends adjustments to lighting and HVAC for efficiency improvement.
- Generates sustainability metrics and carbon footprint analytics.

8. Operational Dashboards and Analytics Tools

- Role-based dashboards for operations, SOC, and event management.
- Insights into performance, energy efficiency, and event analytics.

9. Visitor and Exhibitor Interfaces

- Dedicated web portals for exhibitors to monitor booth traffic, visitor engagement, and analytics in real time.
- Provides visitors with navigation tools, congestion heatmaps, and event updates through accessible interfaces.
- Improve user experience through personalized and data-driven insights.

10. Simulation and Testing Environment

- Controlled environment for validating edge inference, performance, and security responses before deployment.
- Scenario-based stress testing to guarantee operational stability.

2.3 User Classes and Characteristics

SentinaAI will be used by multiple user groups involved in the operation, management, and experience of large convention centers. User classes are differentiated based on their roles, frequency of use and technical expertise.

2.3.1 Primary User Classes

These users are the core operational and analytical stakeholders of the system. Their satisfaction is important for successful system adoption and performance validation.

User Class	Role	Technical Expertise	Access Level & Frequency	Interactions with SentinaAI
Facility Operations Manager	- Oversees the real-time operation of systems such as HVAC, lighting, and occupancy control. - Uses operational dashboards to optimize resource usage.	Moderate technical knowledge of BMS and facility systems.	High frequency; daily use.	Access to operational dashboards, digital twin visualization, and energy optimization tools.
Security Operations Center (SOC) Analyst	- Monitors IoT network behavior, device activity, and threat alerts. - Manages incident response through the security dashboard.	High technical expertise in cybersecurity and monitoring tools.	Continuous; high frequency during events.	Access to threat detection dashboards, alerts, logs, and AI anomaly detection analytics.
Event Administrator	- Configures event layouts, manages visitor zones. - Defines operational parameters during event setup and execution.	Moderate technical expertise and operational management background.	Periodic; high during event scheduling and live operations.	Uses administrative tools to configure zones, assign access rights, and view event analytics.

Exhibitor	Uses analytics to monitor booth traffic, visitor engagement, and performance metrics during events.	Basic; primarily business users.	Periodic; during event days.	Access to exhibitor portal and behavioral analytics tools.
Visitor	Accesses visitor interfaces to navigate the venue, view live occupancy maps, and receive notifications or updates.	Basic; general public.	Intermittent; during event attendance.	Uses the visitor portal and AI assistant for navigation and information queries.

Table 4- User Classes

2.3.2 Secondary User Classes

These users contribute to system setup, maintenance, and continuous improvement but are not the primary users during live event operations.

User Class	Role	Technical Expertise	Access Level & Frequency	Key Interactions with SentinaAI
Maintenance Technician	- Configures and troubleshoots IoT devices and edge nodes. - Ensures system hardware and connectivity reliability.	High technical knowledge of devices networking and protocols.	Regular; based on maintenance schedule.	Access to diagnostic interfaces, system logs, and device calibration tools.
System Administrator	- Manages user accounts, access control, and system configurations. - Ensures uptime and reliability of both edge and cloud services.	High technical and administrative expertise.	Regular; continuous monitoring.	Uses administrative console for configurations, security settings, and backups.
AI / Data Engineer	Develops, trains, and validates AI models for anomaly detection,	Advanced data science and machine learning expertise.	Periodic; as part of model maintenance cycles.	Access to analytics backend and model retraining environment.

	prediction, and optimization.			
Quality Assurance (QA) Engineer	- Tests system performance verifies compliance with requirements. - Ensures reliable operation before deployment.	Moderate to high technical testing knowledge.	Periodic; mainly pre-deployment.	Uses simulation and test environment to validate system functionality.
Executive Management	Reviews performance dashboards, sustainability reports, and key operational KPIs for strategic planning.	Business-level understanding.	Occasional; post-event or monthly review.	Uses high-level dashboards and summary reports for insights and decision-making.

Table 5- Secondary User Classes

2.4 Operating Environment

The system's architecture makes sure that latency-sensitive operations are handled at the edge, while computationally intensive analytics and long-term storage are performed in the cloud.

2.4.1 Technical Environment

1. Architecture split:

- Edge layer:** real-time inference, local automation, short-term buffering, and continued operation during cloud outages.
- Cloud layer:** data aggregation, dashboards, model retraining, long-term storage, and cross-node coordination.

2. Connectivity & security: TLS 1.2+ for all service-to-service traffic; optional site-to-cloud VPN.

3. Containerization: Services are deployed as containerized microservices for portability and scaling.

4. Multi-tenancy (venue-scoped): isolated data/roles per event or zone within one venue.

2.4.2 Hardware Environment

1. Edge Layer (Local Processing):

- Commercially available IoT gateways or embedded controllers (e.g., Raspberry Pi).
- Each node interfaces with connected systems; CCTV, HVAC, lighting, occupancy, and environmental sensors to perform localized inference and control.

2. Edge Data Services:

- Lightweight message broker (MQTT) and embedded store (e.g. SQLite / small time-series DB) for short-term buffering (typ. 24–72 hours) and offline operation; secure, asynchronous cloud sync on reconnection.

3. Cloud Layer (Centralized Analytics):

- Cloud servers host dashboards, long-term databases, AI model training, and orchestration services.
- Deployment will use standard platforms such as AWS, Microsoft Azure, or Google Cloud Platform (TBD).

2.4.3 Software Environment

1. Operating systems:

- **Edge:** Ubuntu Server LTS or Debian-based Linux.
- **Cloud:** Linux hosts running Docker/Kubernetes.
- **Clients:** Modern browsers (Chrome, Edge, Firefox, Safari).

2. Languages & frameworks (indicative):

- Backend in Python/Node.js
- AI with TensorFlow or PyTorch
- Frontend with React or Angular

2.4.4 User Environment

1. **Operations & SOC staff:** role-based web dashboards and the digital twin on standard desktops/laptops.
2. **Event admins, exhibitors, visitors:** access portals and responsive layouts for desktop/tablet/mobile.
3. **Technicians & admins:** secure access to edge/admin consoles (browser or SSH) for configuration and diagnostics.
4. **Engineering staff (cloud/AI/QA):** authenticated access to cloud consoles and CI/CD endpoints.

2.5 Design and Implementation Constraints

As development progresses, additional constraints may arise and will be documented here.

2.5.1 Technical Constraints

1. Must be able to operate using a hybrid edge-cloud architecture, with real-time processing at the edge and analytics in the cloud.
2. Must use existing IoT infrastructure (CCTV, HVAC, lighting, occupancy, and environmental sensors), no new hardware or firmware development.
3. Must support **multi-tenancy** within a single venue for concurrent events or operational zones.

2.5.2 Regulatory and Security Constraints

1. Must comply with data protection and privacy standards, ensuring encryption both in transit and at rest.
2. Must use role-based access control (RBAC) for all user roles and interfaces.
3. Must follow key management practices using approved cloud-native services (AWS KMS, Azure Key Vault, or GCP KMS).
4. Must maintain audit logs for system operations and user actions.
5. All data retention and synchronization policies must follow defined time limits and security rules.

2.5.3 Business and Operational Constraints

1. Must be able to scale horizontally to support increasing numbers of devices or events.
2. Must be maintainable by technical teams with standard training.

2.5.4 User and Interface Constraints

1. Must be accessible through standard web browsers (Chrome, Edge, Firefox, Safari) without additional plugins.
2. Must provide role-based dashboards for facility managers, SOC analysts, exhibitors, and visitors.
3. Must have clear visualization through intuitive dashboards and digital-twin interfaces.
4. Must support accessibility standards for users with disabilities.
5. Must maintain consistent accessibility for users with varying technical proficiency.

2.6 User Documentation

The following user documentation will be developed and delivered alongside the SentinaAI to support onboarding and operations:

1. **User Manual:** Describes platform workflows and usage instructions for facility managers, SOC analysts, event administrators, and exhibitors.
2. **Administrator Guide:** Outlines system configuration, user management, and maintenance procedures for technical and IT staff.
3. **Quick Start / Deployment Guide:** Provides setup and network connection steps for edge and cloud components during initial deployment.
4. **Visitor Guide / Help Pages:** Offers simple guidance embedded within the visitor portal covering navigation live map viewing, FAQ etc.
5. **Online Help and Tooltips:** Context-sensitive support embedded in dashboards and portals to guide users during real-time operation.
6. **Training Material:** Digital training slides to support onboarding for internal teams and event operators.

All documentation will be provided in digital format (PDF and web-based/mobile-based) and written in clear, concise English, following a consistent layout.

2.7 Assumptions and Dependencies

The design and implementation of SentinaAI are based on the following assumptions and dependencies. Any changes to these factors may affect system functionality, performance, or deployment feasibility.

2.7.1 Assumptions

1. The venue will have reliable network infrastructure (LAN/Wi-Fi) to support IoT device connectivity and data transfer between edge and cloud components.
2. The project will use existing IoT devices (CCTV, HVAC, lighting, environmental sensors) that are compatible with standard communication protocols (MQTT, BACnet, Modbus, HTTPS).
3. Commercial cloud services (AWS, Microsoft Azure, or Google Cloud Platform) will be available and approved for deployment.
4. Edge devices will have sufficient computing resources to perform local inference and short-term data buffering.
5. Stakeholders will participate actively in the requirements validation process

2.7.2 Dependencies

1. The platform depends on the availability of stable internet connectivity for synchronization between edge and cloud layers.
2. Operation depends on cloud-based services for centralized analytics, storage, and AI model retraining.
3. The system relies on third-party libraries, frameworks, and APIs (Python, Node.js, TensorFlow, PyTorch, React, etc.) that must remain supported and updated.
4. Security compliance depends on adherence to encryption standards, cloud key management, and institutional data policies.
5. Continued operation and timely availability of power and backup systems for edge devices and network equipment during high traffic/events.

3 System Features and Functional Requirements

This section outlines the functional requirements according to the main system features. Each requirement is assigned a unique identifier (FR-XX) to support clear traceability and reference throughout the development process.

3.1 IoT Device Integration

Management of IoT devices such as CCTV cameras, HVAC controllers, lighting systems, occupancy sensors, and environmental monitors through standardized communication protocols.

Priority: High	
ID	Requirement
FR-01	The system shall support IoT device connectivity using standardized protocols including MQTT, BACnet, Modbus, and HTTPS.
FR-02	The system shall allow discovery and registration of new devices via automated scanning or manual configuration through the admin console.
FR-03	Each registered device shall have a unique ID, metadata record (type, location, function), and associated communication parameters.
FR-04	The system shall support both edge-connected and cloud-connected devices, maintaining consistent data formatting across both.
FR-05	The system shall collect and transmit telemetry data (e.g., temperature, occupancy, energy readings) at configurable intervals.
FR-06	The system shall automatically detect and log device connection losses, retries, and communication failures.
FR-07	The system should support device grouping and tagging by zone, system type, or operational function (e.g., “Hall 1 HVAC,” “Security Cameras”).
FR-08	The system shall implement secure authentication and encryption (TLS 1.2+ or equivalent) for all device-to-edge and edge-to-cloud communication.
FR-09	The system shall provide administrators with a web interface to view, enable, disable, or reassign devices.
FR-10	The system shall maintain local buffering of telemetry on the edge node during temporary network outages, resending data when connectivity is restored.
FR-11	The system should support over-the-air (OTA) configuration updates for compatible devices without manual intervention.
FR-12	The system may provide diagnostic tools to test device connectivity, latency, and data integrity from the management console.
FR-13	The system shall log all device onboarding, status changes, and communication events for auditing and traceability.
FR-14	The system may support third-party device simulation for development or testing purposes within the simulation environment.
FR-15	The system shall make sure that device integration processes do not exceed defined latency thresholds for real-time data capture (≤ 2 seconds for critical sensors).

Functional Requirements 1- IoT Integration

3.2 Edge Node and Local Intelligence

Real-time decision-making, automation, and local anomaly detection at the edge layer to reduce latency and continued operation during cloud disconnection.

Priority: High	
ID	Requirement
FR-16	Each edge node shall process incoming sensor and device data locally for real-time inference and control decisions.
FR-17	Edge nodes shall perform anomaly detection on local data streams using deployed ML models without relying on the cloud.
FR-18	Buffered data shall synchronize automatically with the cloud once connectivity is restored, ensuring no data loss or duplication.
FR-19	Edge nodes shall run modular microservices for telemetry ingestion, analytics, and rule-based automation.
FR-20	The system shall allow remote deployment and updating of edge AI models and configuration packages from the cloud.
FR-21	Edge nodes should support prioritization of mission-critical tasks (e.g., security feeds) over non-critical telemetry during overload.
FR-22	The system shall encrypt all edge-to-device and edge-to-cloud communications using TLS 1.2 or higher.
FR-23	Each edge node shall store logs of local events, inference results, and control actions for later analysis.
FR-24	Edge nodes should include a local rules engine that supports condition-action logic (e.g., “if CO ₂ > 900 ppm then increase airflow”).
FR-25	The system may allow dynamic re-allocation of processing tasks across multiple edge nodes for load balancing.
FR-26	The system should monitor edge resource usage (CPU, memory, storage) and send health reports to the cloud.
FR-27	Edge nodes shall continue operating autonomously when disconnected from the cloud and resync automatically on reconnection.
FR-28	The framework may support simulation of edge logic and model inference within the test environment for validation before deployment.
FR-29	Edge nodes shall implement secure boot and code-signing verification to prevent unauthorized firmware execution.

Functional Requirements 2- Edge Intelligence

3.3 Cloud Coordination and Data Analytics

Centralized data aggregation, AI model retraining, and analytics visualization across all edge and venue systems.

Priority: High	
ID	Requirement
FR-30	The system shall aggregate telemetry, logs, and performance data from all edge nodes into a unified cloud data repository.
FR-31	The cloud platform shall provide centralized orchestration and monitoring of all connected edge nodes and IoT devices.
FR-32	The system shall perform large-scale data analytics, including trend detection, forecasting, and anomaly correlation.
FR-33	The cloud layer shall host the machine learning pipelines used for model retraining and redeployment to edge nodes.
FR-34	The system shall synchronize metadata, configuration changes, and model updates between edge and cloud in near real time.
FR-35	The cloud shall provide secure multi-tenant data isolation for different events or venue zones within the same deployment.
FR-36	The system shall ensure data integrity through checksums, transaction validation, and time-synchronized versioning.
FR-37	The cloud shall maintain long-term storage of historical data for analytics, with configurable retention policies.
FR-38	The system should use scalable containerized services (e.g., Kubernetes or Docker) for deployment flexibility and load balancing.
FR-39	The cloud shall expose internal APIs for communication with dashboards, AI modules, and the anomaly detection framework.
FR-40	The system shall not expose public APIs or third-party data access in this project phase.
FR-41	The system shall support advanced query and filtering options for time-series and event data across nodes.
FR-42	The system shall ensure encryption of all data in transit and at rest using AES-256 and TLS 1.2+ standards.
FR-43	The platform should provide visualization and reporting tools for global trends, cross-event analytics, and sustainability metrics.
FR-44	The cloud should support alert correlation logic, linking multiple anomalies or faults into unified incident reports.
FR-45	The system may integrate automated performance optimization for data pipelines, such as dynamic caching or batching.
FR-46	The system may generate AI-driven insights or recommendations for energy optimization, crowd management, or maintenance planning.
FR-47	The cloud shall maintain high availability with redundancy across multiple availability zones within the selected provider (AWS, Azure, or GCP).
FR-48	The system shall log all orchestration actions, model updates, and data synchronization events for traceability.

Functional Requirements 3- Cloud and Data Analytics

3.4 AI-Driven Anomaly and Security Detection

Detects cyber threats, device faults, and operational anomalies across edge and cloud layers using AI models for early warning and response.

Priority: High	
ID	Requirement
FR-49	The system shall continuously monitor IoT network traffic, device telemetry, and user activity for anomalies and irregular patterns.
FR-50	The framework shall employ AI/ML models to detect operational or security anomalies, including abnormal traffic spikes, sensor data deviations, or unauthorized access attempts.
FR-51	The system shall classify anomalies by severity and type (e.g., device fault, network intrusion, energy deviation, or environmental outlier).
FR-52	The framework shall generate real-time alerts and route them to the appropriate operational dashboards and SOC interface.
FR-53	The system shall log each anomaly event with metadata including timestamp, device ID, severity level, probable cause, and resolution status.
FR-54	The system shall provide configurable detection thresholds and alert policies that can be tuned by administrators.
FR-55	The framework should use multi-layer detection techniques, statistical baselines, behavior modeling, and rule-based filters to minimize false positives.
FR-56	The system shall correlate anomalies across multiple edge nodes and time windows to identify coordinated or recurring events.
FR-57	The framework should integrate with the digital twin to visually highlight affected devices and systems in real time.
FR-58	The system shall allow authorized users to acknowledge and dismiss alerts via the dashboard interface.
FR-59	The framework shall maintain historical anomaly data for trend analysis, mean-time-to-detect (MTTD), and mean-time-to-resolve (MTTR) metrics.
FR-60	The system shall enable administrators to simulate anomaly scenarios in the testing environment to validate detection performance.
FR-61	The framework should adaptively retrain its AI models based on feedback from verified anomaly cases.
FR-62	The system shall enforce encryption and authentication for all anomaly data exchanged between edge, cloud, and dashboards.
FR-63	The framework should support automated responses for specific anomaly types, such as isolating a compromised device or restarting a failed service.
FR-64	The system may provide anomaly correlation analytics, identifying links between system performance, visitor load, and environmental changes.
FR-65	The system may generate periodic anomaly trend reports summarizing event frequency and overall system health.
FR-66	The framework shall ensure compliance with internal data governance policies and shall not expose detection data via external APIs.
FR-67	The system shall be capable of operating under degraded network conditions by maintaining localized anomaly logs until synchronization is restored.
FR-68	The framework should provide a “threat learning mode” that enables model improvement using synthetic data from simulation tests.

Functional Requirements 4- Anomaly Detection

3.5 Operational Dashboards and Visualization Tools

Role-based dashboards for operations, security, and sustainability metrics with real-time data visualization.

Priority: High	
ID	Requirement
FR-69	The system shall provide a role-based dashboard accessible via secure web browsers for all authorized user types.
FR-70	The dashboards shall display live data streams from IoT devices, edge nodes, and cloud analytics modules in real time (≤ 5 s refresh).
FR-71	The system shall include configurable widgets and panels for key domains: security, operations, sustainability, and visitor analytics.
FR-72	Each user shall view only data permitted under their assigned access level (e.g., SOC analyst vs. exhibitor).
FR-73	The dashboards shall include alert panels displaying active anomalies, security threats, and equipment faults.
FR-74	The system shall allow users to filter and search dashboard data by time and zone
FR-75	The dashboards should support visual elements such as heatmaps, trend charts, gauges, and timeline graphs.
FR-76	The system shall allow quick drill-down from high-level KPIs to detailed device-level or event-level metrics.
FR-77	The system shall integrate with the Digital Twin module to visualize spatial data, occupancy, and energy flow.
FR-78	The dashboards should include historical playback for reviewing system states or incidents over selectable time ranges.
FR-79	The system shall include an audit trail logging all dashboard user actions (login, view, command execution).
FR-80	The dashboards should support customizable layouts so each role can tailor their workspace.
FR-81	The system shall ensure all data visualizations auto-update when new data is received from edge or cloud nodes.
FR-82	The dashboards should support KPI scorecards summarizing operational efficiency, sustainability, and safety metrics.
FR-83	The system shall provide secure HTTPS-based access and enforce multi-factor authentication for dashboard logins.
FR-84	The dashboards may include quick-action controls for authorized users (e.g., reset device, acknowledge alert).
FR-85	The system should include data-export options (PDF, CSV) for reports and presentations.
FR-86	The dashboards may include AI-generated summaries or insights provided through integration with the AI Assistant.

Functional Requirements 5- Dashboards and Visualization

3.6 Digital Twin Environment

3D virtual representation of the venue that visualizes occupancy, energy flow, and system performance in real time.

Priority: Medium	
ID	Requirement
FR-87	The system shall maintain a real-time 3D representation of the venue, synchronized with IoT data from sensors, HVAC, lighting, and occupancy systems.
FR-88	The digital twin shall receive updates from edge nodes and the cloud layer within 5 seconds of telemetry generation.
FR-89	The digital twin shall visually represent system status through color-coded overlays for occupancy, temperature, and device health.
FR-90	The system shall allow users to navigate between different spatial levels (zones, halls, floors) and view associated metrics.
FR-91	Each object or zone within the digital twin shall be linked to its metadata record (e.g., device type, ID, location, function).
FR-92	The digital twin should allow filtering of displayed data layers (e.g., HVAC, lighting, security, energy).
FR-93	The system shall support playback of historical data, allowing visualization of occupancy or environmental changes over time.
FR-94	The digital twin should allow users to overlay anomaly data and system alerts from the AI Detection Framework.
FR-95	The system shall store and display simulation results generated in the testing environment for validation purposes.
FR-96	The digital twin should display contextual KPIs (e.g., energy use, CO ₂ levels, comfort index) for each zone.
FR-97	The system shall log all user interactions and view changes for traceability and analytics.
FR-98	The digital twin may allow exporting static snapshots or short animations for reporting purposes.
FR-99	The system may support predefined scene views (e.g., “Energy Mode,” “Crowd Density,” “Security Map”) for quick context switching.
FR-100	The system shall comply with all performance and visualization limits defined for web-based dashboards.
FR-101	The system should optimize rendering performance to maintain frame rates above 25 FPS during typical venue operations.
FR-102	The system may integrate with the AI Assistant to support contextual queries (e.g., “Highlight zones exceeding 80% occupancy”).

Functional Requirements 6- Digital Twin

3.7 AI Assistant and Interaction Layer

Natural-language interfaces for simplified system control and data access through text or voice.

Priority: Medium	
ID	Requirement
FR-103	The system shall embed a natural-language assistant accessible from all web dashboards and user portals.
FR-104	The assistant shall interpret user queries and map them to predefined commands, data requests, or actions.
FR-105	The system shall allow users to query real-time or historical data
FR-106	The assistant shall enforce access control, ensuring responses are limited to the user's authorization level.
FR-107	The assistant shall support context-aware dialogue, allowing users to issue follow-up questions without restating context.
FR-108	The system shall retrieve requested information from analytics, anomaly detection, or digital twin modules.
FR-109	The assistant should recognize synonyms, paraphrases, and partial commands to improve interaction flexibility.
FR-110	The assistant shall log all interactions (query, user role, timestamp, response type) for auditing and model improvement.
FR-111	The system shall display assistant responses in structured formats such as tables, text summaries, or linked dashboards.
FR-112	The assistant should handle clarification prompts for incomplete or ambiguous user inputs.
FR-113	The assistant may generate natural-language summaries (e.g., "Today's security events" or "Energy usage trends this week").
FR-114	The system shall prevent execution of unauthorized commands or unsafe actions through strict input sanitization and access validation.
FR-115	The assistant shall maintain session continuity for the user during a dashboard session.
FR-116	The system shall make sure encryption of all assistant queries and responses transmitted between the browser, cloud, and edge services.

Functional Requirements 7- AI Assistant

3.8 Sustainability and Energy Optimization Module

Monitors and optimizes energy consumption and forecasting sustainability performance.

Priority: High	
ID	Requirement
FR-117	The system shall continuously collect energy consumption data from HVAC, lighting, and environmental sensors.
FR-118	The module shall calculate real-time energy usage metrics at device, zone, and venue levels.
FR-119	The system shall generate efficiency KPIs such as energy-per-visitor, energy-per-zone, and total daily consumption.
FR-120	The system shall use AI models to predict short-term energy demand based on occupancy and environmental conditions.
FR-121	The module should identify and flag inefficiencies such as overcooling, lighting overuse, or sensor malfunctions.
FR-122	The system shall recommend optimization actions (e.g., adjust HVAC schedule, dim lighting) based on predictive analytics.
FR-123	The module should track sustainability indicators such as CO ₂ emissions, energy savings, and operational efficiency.
FR-124	The system shall provide a visual sustainability dashboard showing energy flow, efficiency trends, and sustainability scores.
FR-125	The module should correlate energy usage with occupancy and event activity data to support data-driven planning.
FR-126	The system shall store historical energy and sustainability data for trend and performance analysis.
FR-127	The module may trigger alerts or notifications when energy consumption exceeds predefined thresholds.
FR-128	The system should support simulation of energy-saving strategies using the digital twin for predictive validation.
FR-129	The module shall ensure data integrity and synchronization between edge and cloud analytics for all energy data.
FR-130	The system should generate automated sustainability reports summarizing monthly and event-based performance.
FR-131	The module may integrate weather forecast data to refine energy demand predictions.
FR-132	The system shall comply with organizational sustainability policies and shall not expose data externally in this phase.

Functional Requirements 8- Sustainability

3.9 Visitor and Exhibitor Interfaces

Web-based interfaces for visitors and exhibitors to access heatmaps and booth analytics.

Priority: Medium	
ID	Requirement
FR-133	The system shall provide separate interfaces for visitors and exhibitors, accessible through secure authentication.
FR-134	The visitor interface shall display real-time crowd density maps, event schedules, and navigation guidance within the venue.
FR-135	The exhibitor interface shall display booth analytics, including visitor count, dwell time, and engagement statistics.
FR-136	The interfaces shall maintain strict data segmentation, exhibitors can only view analytics related to their assigned booths.
FR-137	The system shall support role-based permissions distinguishing visitor, exhibitor, and admin access levels.
FR-138	The visitor portal should provide interactive maps linked to the digital twin for route planning and congestion alerts.
FR-139	The exhibitor dashboard should include historical comparison charts (e.g., current hour vs. previous day or previous event).
FR-140	The system shall anonymize all visitor data shown in exhibitor analytics to comply with data privacy requirements.
FR-141	The visitor interface shall allow querying event-related information (e.g., “nearest food court,” “sessions starting soon”).
FR-142	The exhibitor portal should support export of booth analytics reports (PDF/CSV).
FR-143	The system may allow exhibitors to view engagement heatmaps derived from crowd-tracking sensors near their booths.
FR-144	The visitor interface may integrate contextual notifications (e.g., “Hall 5 nearing full capacity”) for crowd management.
FR-145	The interfaces should allow access to AI Assistant queries for quick insights (e.g., “Which zones are busiest now?”).
FR-146	The system shall log all portal interactions (user type, time, query, accessed data) for analytics and security auditing.

Functional Requirements 9- User Interfaces

3.10 Behavioral and Operational Analytics

Insights on visitor behavior, exhibitor performance, and operational efficiency using aggregated data.

Priority: High	
ID	Requirement
FR-147	The system shall collect, aggregate, and process behavioral data including visitor dwell time, zone transitions, and engagement patterns.
FR-148	The analytics engine shall correlate sensor data, camera metadata, and entry logs to derive operational insights.
FR-149	The system shall calculate performance indicators such as average dwell time per zone, occupancy efficiency, and flow congestion index.
FR-150	The module shall support real-time dashboards for visualizing crowd density, movement paths, and engagement heatmaps.
FR-151	The analytics engine should apply statistical and AI models to identify recurring behavior patterns or inefficiencies.
FR-152	The system shall anonymize all personal and sensor data before behavioral analysis for privacy compliance.
FR-153	The analytics framework shall allow filtering and comparison of behavioral data across events, days, or time intervals.
FR-154	The system should provide predictive analytics estimating expected crowd distribution and resource demand during events.
FR-155	The analytics shall generate periodic reports summarizing visitor engagement, booth performance, and service efficiency.
FR-156	The system should support configurable thresholds and alerting for abnormal crowd density or under-utilized zones.
FR-157	The module may visualize relationships between crowd flow, energy consumption, and operational cost for cross-domain optimization.
FR-158	The analytics engine shall synchronize computed results with the cloud data warehouse for historical trend analysis.
FR-159	The system shall allow exporting analytics results (CSV, PDF) for stakeholders and management reports.
FR-160	The module may integrate with the AI Assistant for contextual queries (e.g., “Which halls had the highest dwell time yesterday?”).
FR-161	The system shall maintain data quality standards by validating timestamps and sensor inputs.
FR-162	The module should enable benchmarking between different events to measure long-term operational improvement.

Functional Requirements 10- Behavioral and Operational Analytics

3.11 Data Governance and System Reliability

Data integrity and high availability through redundancy, fault tolerance, secure storage practices etc.

Priority: High	
ID	Requirement
FR-163	The system shall enforce end-to-end encryption for all data in transit and at rest using TLS 1.2+ and AES-256 standards.
FR-164	The system shall implement role-based access control (RBAC) to restrict data and feature access based on user roles.
FR-165	The platform shall maintain audit logs of all authentication events, configuration changes, and administrative actions.
FR-166	The system shall replicate critical data across multiple availability zones or redundant storage nodes to prevent data loss.
FR-167	The system shall include automated health checks to detect service failures, performance degradation, or unreachable nodes.
FR-168	The platform shall support automatic fail-over and recovery for key services (e.g., databases, edge gateways, dashboards).
FR-169	The system shall maintain integrity validation using checksums, version tracking, and transaction consistency checks.
FR-170	The platform should provide configurable backup intervals for databases and operational logs (e.g., daily, hourly).
FR-171	The system shall make sure all backups are encrypted and stored securely within the same cloud region or an approved replica.
FR-172	The system shall detect and alert administrators in the event of corruption, unauthorized access, or configuration drift.
FR-173	The platform should provide real-time system health dashboards displaying uptime, latency, and resource usage metrics.
FR-174	The system shall enforce strict authentication policies including MFA for administrative and operator accounts.
FR-175	The system should support configurable data-retention policies to meet venue or regulatory requirements.
FR-176	The platform may employ anomaly detection to monitor for irregular data-access behavior or failed authentication attempts.
FR-177	The system shall maintain transactional logs to ensure traceability between raw sensor data, processed analytics, and reported metrics.
FR-178	The system should support distributed locking or consensus mechanisms to avoid data conflict between cloud and edge nodes.
FR-179	The system shall ensure time synchronization between edge and cloud components for consistent event sequencing.
FR-180	The system may allow periodic internal audits or verification scripts to validate data lineage and compliance adherence.

Functional Requirements 11- Data Governance and Reliability

3.12 Simulation and Testing Environment

Controlled environment for testing AI models, system stability, and security responses before deployment.

Priority: Medium	
ID	Requirement
FR-181	The system shall provide an isolated simulation environment separate from production systems for safe testing and validation.
FR-182	The environment shall support testing of AI models for edge inference, anomaly detection, and automation logic.
FR-183	The system shall allow users to simulate IoT device data streams to evaluate performance under various load conditions.
FR-184	The simulation shall include configurable parameters for latency, packet loss, and sensor frequency to mimic real-world variability.
FR-185	The environment should enable testing of system recovery and synchronization after simulated network disconnections.
FR-186	The system shall provide tools to replay historical datasets for model validation and analytics benchmarking.
FR-187	The simulation shall log all test scenarios, configurations, and outcomes for reproducibility and auditability.
FR-188	The system should provide visualization of simulated data through the digital twin to assess behavioral accuracy.
FR-189	The environment shall enable administrators to test new software releases, patches, or configuration changes before deployment.
FR-190	The system should allow side-by-side comparison of simulated vs. live results to evaluate inference accuracy.
FR-191	The environment shall not impact or overwrite live production data during any simulation.
FR-192	The system may support automated regression testing scripts for continuous integration workflows.
FR-193	The environment may integrate performance benchmarking tools to measure latency, throughput, and resource utilization.
FR-194	The system shall allow export of test reports summarizing results, anomalies detected, and system performance metrics.
FR-195	The environment should provide configurable test scenarios for security validation, including simulated DDoS or unauthorized access attempts.
FR-196	The system shall support model retraining workflows triggered by simulation feedback for performance improvement.
FR-197	The simulation environment shall comply with all security, access control, and data governance policies applied in production.

Functional Requirements 12- Simulation and Testing

Note: [Appendix A](#) contains the full use-case descriptions. Only the primary, high-value scenarios have been included to support understanding of the functional requirements.

4 External Interface Requirements

4.1 User Interfaces

ID	Requirement
UI- 01	The interface shall display anomalies, specify the industrial protocols, and identify the sequence integrity failures and associated contexts.
UI- 02	The dashboard shall visualize occupancy and energy usage and other metrics with interactive heatmaps.
UI- 03	The dashboards shall allow customized, role-based views for various users such as analysts and admins.
UI- 04	Dashboards shall allow filtering and drilldowns by anomaly type and time.
UI- 05	Dashboards shall allow secure triggering of edge node commands.
UI- 06	The dashboard shall allow users to access and view anomaly logs and other data for analysis.
UI- 07	The dashboard shall present real-time statuses of the automated responses for incidents using colour-coded indicators such as red, yellow, or green, depending on the severity of the incidents
UI- 08	The interface shall enforce RBAC (role-based access control) to ensure alerts are prioritized based on the users.
UI- 09	All alerts should display the AI model's confidence score to aid analysts in assessing the severity of alerts alongside colour-coded indicators.
UI- 10	The interface shall provide graphical trend analysis showing the type of anomalies and their frequencies over time.
UI- 11	The dashboard shall allow filtering and searching audit logs by their zone IDs, response tiers, device metadata, and containment periods.
UI- 12	The dashboard shall display a real-time health check status of the connection continuously between the edge node and the network switch.
UI- 13	The dashboard must provide a secure session and authentication for all users.
UI- 14	The dashboard shall allow export interfaces to generate various types of reports in PDFs and spreadsheets.
UI- 15	The system shall provide users with a browser-based dashboard for real-time monitoring of edge nodes and venues.
UI- 16	The UI shall display data through widgets such as graphs, maps, and such that can be customized and rearranged by authorized users.
UI- 17	The UI shall stick to responsible web design standards to ensure compatibility with various devices and control room displays.
UI- 18	All error messages shall include timestamps, system codes, and user-friendly descriptions to support troubleshooting.
UI- 19	The UI shall provide screens where each screen contains a consistent help icon linked to user guide and tutorials.
UI- 20	The SOC dashboard should refresh automatically every few seconds without the need for human/manual intervention.
UI- 21	The SOC dashboard shall allow the analysts to classify records as a True positive or a false positive for the AI training dataset.

UI- 22	The visitor portals shall display optimized routes on the maps, which shall update in real time based on live data from the navigation system.
UI- 23	The system shall enforce a timeout policy that terminates the active session after 20 continuous minutes of inactivity and shall prompt the user for re-authentication.
UI- 24	The dashboard shall provide both dark and light colour themes to accommodate personal preferences, lighting conditions, and ensure optimal readability.
UI- 25	The interface shall follow accessibility standards to accommodate People of Determination (PoD).
UI- 26	The interface shall scale properly on various resolutions used for monitoring
UI- 27	The dashboard shall provide push notifications for critical anomaly detections
UI- 28	The UI shall display alerts for failed API calls or communication issues with edge nodes
UI- 29	The interface shall support drag and drop configuration for widgets and layout customizations
UI- 30	The interface shall provide a notification center that summarizes alerts and updates
UI- 31	The UI shall integrate a feedback panel allowing users to report interface issues directly to system administrators
UI- 32	The interface shall provide users with an integrated AI assistant that can be accessed from the dashboard via icons.
UI- 33	The AI assistant shall provide conversational query capabilities, allowing users to ask questions and receive conversational answers
UI- 34	The AI assistant shall provide suggestions based on context and user roles.
UI- 35	The AI assistant shall make the users aware of issues with connectivity or telemetry that are affecting its ability to answer queries.

External Interface 1- UI Requirements

4.2 Hardware Interfaces

ID	Requirement
HI-01	The system shall provide a secure interface for CCTV devices using RTSP protocols for live streaming and triggers that would be event-based.
HI-02	The system shall allow resolution adjustment, motion-triggered captures, and other bidirectional commands to CCTV cameras.
HI-03	The system shall allow interfacing with sensors using MQTT and other protocols to capture motion, density, and presence.
HI-04	The system shall allow interfacing with HVAC systems via Modbus to adjust variables such as temperature and humidity.
HI-05	The system shall interface with sensors using MQTT and transmit raw and processed data.
HI-06	The system shall provide a secure MQTT interface to enable occupancy-based adjustments to smart lighting controllers.
HI-07	The system shall use locally stored credentials to enable secure authentication and authorization for devices.
HI-08	The system shall store metadata from devices such as ID, Location, etc., in the edge node device registry DB.

HI-09	The systems shall allow synchronization automatically with the edge node after buffering telemetry during network disruptions, once connectivity resumes.
HI-10	The system shall yield error reporting interfaces, including sensor failures, communication problems, and calibration drift, to edge nodes.
HI-11	The systems shall support the onboarding of new devices using protocol adapters, supporting plug-and-play.
HI-12	The AI assistant shall automatically detect unavailable hardware, such as speakers or microphones, and disable related functionalities
HI-13	The AI assistant shall interact with sensors and other devices indirectly through the edge node hardware interface
HI-14	The system shall be compatible with industry-grade gateways that would be capable of running Linux-based agents.
HI-15	The system shall maintain consistent low-latency communication between hardware devices and edge nodes
HI-16	The system shall provide monitoring of power usage, uptime, and downtime of hardware for edge devices.
HI-17	The system shall ensure firmware update interfaces for devices to be available through secure OTA processes.

External Interface 2- Hardware

4.3 Software Interfaces

ID	Requirement
SI-01	Consistent and real-time telemetry from devices shall be received by the edge node
SI-02	The edge node shall perform AI inference using locally cached models
SI-03	The edge node shall execute automation rules and send commands based on pre-defined thresholds to lighting, HVAC, or alerts.
SI-04	The edge node shall of all telemetry, anomalies, and other actions to the log buffer database for future synchronizations.
SI-05	The edge node shall provide containerized microservices for AI inference and allow independent restarts using telemetry processing.
SI-06	The edge node shall expose local APIs for dashboards to allow querying.
SI-07	The edge node should maintain autonomous operations using cached models when cloud connectivity is unavailable.
SI-08	The dashboard shall allow future integrations of new device types with minimal interface redesigns.
SI-09	The cloud shall validate and filter all incident logs using classification tags before transferring data to the AI training dataset.
SI-10	The AI assistant shall consume APIs exposed by the dashboard to retrieve logs and telemetry, as well as anomaly data.
SI-11	The AI assistant shall integrate with the NLP engine hosted on the cloud for NLU.
SI-12	The AI extension shall support plugin-based extension APIs for scalability and future integrated capabilities.
SI-13	The system shall integrate with a SQL or NoSQL DB to store metadata, telemetry, and AI conversation logs.

SI-14	The system shall utilize API gateways to manage request routing and rate limiting across microservices.
SI-15	The system shall support integration with third-party analytics engines via REST for model retraining.
SI-16	The edge software shall maintain compatibility with Docker Compose/Kubernetes
SI-17	The system shall maintain version control for deployed AI models, ensuring compatibility.

External Interface 3- Software

4.4 Communication Interfaces

ID	Requirement
CI-01	The edge node shall communicate with the cloud layers via HTTPS or other suitable protocols over TLS, for model updates, synchronization of telemetry, and for reporting.
CI-02	The interface shall issue emergency/urgent notifications if an automatic rollback was initiated at the edge node, also including the version reverted and the reason.
CI-03	The AI assistant shall communicate with cloud-based NLP and LLM through secure HTTPS connections.
CI-04	The AI assistant shall encrypt all queries and responses to ensure privacy
CI-05	The assistant shall log all communication requests with timestamps.
CI-06	The system shall support data transmission that's bandwidth adaptive.
CI-07	The communication layer shall comply with IPv4, IPv6 standards.
CI-08	The system shall log all communication packets for diagnostics.
CI-09	The system shall comply with industrial communication standards
CI-10	The AI assistant shall support synchronous and asynchronous messaging with an NLP backend.
CI-11	All telemetry queries and AI responses shall be encrypted end-to-end
CI-12	The system shall support fallback to edge caching if cloud communication fails
CI-13	Transmission failures shall be logged by communication interfaces with timestamps and be tried again automatically.
CI-14	The edge node shall wait for a confirmation from the cloud logs that the Tier 3 containment action has been received and archived before other incident cleanups.

External Interface 4- Communication

5 Other Nonfunctional Requirements

5.1 Performance Requirements

5.1.1 Data Processing Requirements

ID	Requirement
NFR-01	The system shall process data to dashboards within 2 seconds of the data value updating to make sure real-time monitoring.
NFR-02	The system shall execute AI model inference within 500 milliseconds per event to maintain responsiveness during anomaly detection.

Nonfunctional Requirements 1- Performance (Data Processing)

5.1.2 Throughput Requirements

ID	Requirement
NFR-03	The system shall support a minimum of 50 concurrently active IoT devices during the demonstration phase.
NFR-04	The system shall be able to process no less than 500 MQTT messages a second under normal operational load.

Nonfunctional Requirements 2- Performance (Throughput)

5.1.3 Stability and Utilization Requirements

ID	Requirement
NFR-05	The system shall maintain CPU utilization below 80% on edge devices under full load conditions to prevent thermal throttling.
NFR-06	The system shall maintain operation for at least 5 minutes in offline mode if the cloud connection is interrupted, ensuring temporary autonomy.
NFR-07	The system shall not lose more than 0.1% of the telemetry data in case of a power outage or restart.

Nonfunctional Requirements 3- Performance (Stability and Utilization)

5.1.4 AI Requirements

ID	Requirement
NFR-07	The AI model shall maintain 85% of precision during anomaly detection.
NFR-08	The AI model shall maintain a false positive percentage under 15% to prevent unnecessary alerts.

Nonfunctional Requirements 4- Performance (AI)

5.1.5 UI Experience Performance Requirements

ID	Requirement
NFR- 10	The dashboard shall load within 3 seconds of launching (under normal conditions)
NFR- 11	The visual twin interface shall render the real-time data at a minimum of 24/30FPS (frames per second) for easy viewing.
NFR- 12	The system shall support at least three dashboard users concurrently without lag.
NFR- 13	The chatbot UI shall respond to queries made by analysts within 3 seconds of making mentioned queries.
NFR- 14	The UI of the visual twin shall navigate within less than a second of the user's input.

Nonfunctional Requirements 5- Performance (UI Experience)

5.2 Security Requirements

5.2.1 Authentication and Authorization

ID	Requirement
NFR- 15	The system shall implement MFA for administrative access to the monitoring dashboard.
NFR- 16	The system shall enforce a strong password policy for all accounts.
NFR- 17	The system shall implement RBAC to separate operator and admin privileges.
NFR- 18	The system shall log all authentication and access attempts.
NFR- 19	The system shall automatically lock accounts after a number of failed login attempts.
NFR- 20	The system shall maintain secure session management by implementing automatic logout after a period of inactivity.

Nonfunctional Requirements 6- Security (Authentication and Authorization)

5.2.2 Data Protection and Privacy

ID	Requirement
NFR- 21	The system shall encrypt data in transit with TLS.
NFR- 22	The system shall encrypt sensitive data at rest with AES-256.
NFR- 23	The system shall securely store credentials and ensure passwords are not saved in plain text.
NFR- 24	The system shall anonymize people counting data and not store any facial identification or personal identification data.
NFR- 25	The system shall collect only the minimum amount of data required to operate therefore abiding by the data-minimization principle.

Nonfunctional Requirements 7- Security (Data Protection)

5.2.3 Network and IoT Security

ID	Requirement
NFR- 26	The system shall authenticate all IoT devices communicating with MQTT broker.
NFR- 27	The system shall restrict unauthorized publishing and subscribing to MQTT topics
NFR- 28	The MQTT broker and Node-RED services shall use TLS secured communication.
NFR- 29	The system shall ensure least privilege network access by restricting device to device communication unless required.

Nonfunctional Requirements 8- Security (Network and IoT)

5.3 Operational Requirements

5.3.1 System Availability and Reliability

ID	Requirement
NFR- 30	The system shall maintain an uptime of at least 99.9% under normal operating conditions.
NFR- 31	The system shall automatically restart or recover from minor software or hardware faults without data loss.
NFR- 32	The system shall guarantee uninterrupted operation of monitoring components during routine maintenance.
NFR- 33	The system shall support redundant network paths to ensure continuous communication with IoT devices.

Nonfunctional Requirements 9- Operational (Availability and Reliability)

5.3.2 Performance & Scalability

ID	Requirement
NFR- 34	The system shall process and display alerts within 2 seconds of anomaly detection.
NFR- 35	The system shall scale horizontally by adding edge nodes without requiring system downtime.
NFR- 36	The system shall maintain acceptable performance even under peak load conditions or partial outages.

Nonfunctional Requirements 10- Operational (Performance and Scalability)

5.3.3 Maintenance and Support

ID	Requirement
NFR- 37	The system shall schedule automatic backups of configuration and log data daily.
NFR- 38	The system shall support quick rollback to previous configurations in case of update failures.
NFR- 39	The system shall provide diagnostic tools for troubleshooting and performance monitoring.

Nonfunctional Requirements 11- Operational (Maintenance and Support)

5.3.4 Security and Compatibility

ID	Requirement
NFR- 40	The system shall ensure encrypted communication between all IoT nodes and the central monitoring server.
NFR- 41	The system shall comply with organizational cybersecurity standards and access control policies.

Nonfunctional Requirements 12- Operational (Compatibility)

5.3.5 Documentation and Training

ID	Requirement
NFR- 42	The system shall include comprehensive documentation for installation, configuration, and maintenance.
NFR- 43	The system shall maintain version-controlled documentation updated with each system release.

Nonfunctional Requirements 13- Operational (Documentation)

5.4 Political and Legal Requirements

5.4.1 Regulatory & Compliance Requirements

ID	Requirement
NFR- 44	The system shall comply with UAE Federal Decree-Law No. 45 of 2021 on Personal Data Protection (PDPL).
NFR- 45	The system shall honour data-subject rights including access, correction, deletion, and objection.
NFR- 46	The system shall align with ISO/IEC 27001:2022 information-security controls.
NFR- 47	The system should follow NIST Cybersecurity Framework (CSF) best practices.
NFR- 48	The system should follow ISO/IEC 27017 and ISO/IEC 27018 for cloud security and PII protection.
NFR- 49	The system shall comply with PDPL cross-border data-transfer rules.

Nonfunctional Requirements 14- Regulatory and Compliance

5.4.2 Logging Requirements

ID	Requirement
NFR- 50	The system shall maintain tamper-proof audit logs.
NFR- 51	Audit logs shall be protected with integrity mechanisms.
NFR- 52	The system shall support breach-notification processes.

Nonfunctional Requirements 15- Logging

5.4.3 Intellectual Property and Software Licensing Requirements

ID	Requirement
NFR- 53	All third-party libraries shall comply with their open-source licenses.

Nonfunctional Requirements 16- Intellectual and Licensing

5.4.4 Hardware and IoT Compliance Requirements

ID	Requirement
NFR- 54	Edge hardware shall comply with IEC 62368-1 safety standards.
NFR- 55	IoT wireless modules shall comply with applicable IEEE and Bluetooth standards.

Nonfunctional Requirements 17- IoT Compliance

5.4.5 AI Governance and Ethics Requirements

ID	Requirement
NFR- 56	AI components should follow global AI ethics principles.
NFR- 57	The system shall guarantee AI decisions avoid harmful bias and enable human oversight.
NFR- 58	The system should support explainability for AI decisions affecting operations.
NFR- 59	The system should support multi-jurisdictional compliance readiness aligned with GDPR.

Nonfunctional Requirements 18- AI Governance and Ethics

5.5 Other Constraints

5.5.1 Training Requirements

These constraints define the level of training and guidance required for administrators and staff to effectively operate and maintain the system.

ID	Requirement
OC- 1	The system shall require minimal training for administrators to operate the monitoring dashboard and incident response features efficiently.

Other Constraints 1- Training

5.5.2 Data Handling Constraints

These constraints ensure that the system adheres to security and data management standards to protect sensitive information.

ID	Requirement
OC- 2	The system shall not store or process any facial recognition data or personally identifiable information (PII).
OC- 3	The system shall comply with organizational data retention and disposal policies for IoT and security logs.
OC- 4	The system shall ensure that all stored data is encrypted at rest and in transit following TLS standards.

Other Constraints 2- Data Handling

5.5.3 Deployment Constraints

These constraints define the operational limits and environmental conditions under which the system must function effectively.

ID	Requirement
OC- 5	The system shall operate efficiently on standard IoT and edge infrastructure without the need for specialized hardware.
OC- 6	The system shall be compatible with existing network configurations

Other Constraints 3- Deployment

Appendix A: Glossary

Term	Definition
IoT	Internet of Things: a network of interconnected devices that communicate data.
AI	Artificial Intelligence
MFA	Multi-Factor Authentication: a security method requiring two or more verification factors before granting access.
RBAC	Role-Based Access Control: an access control model where permissions are assigned according to user roles.
TLS	Transport Layer Security: a cryptographic protocol that encrypts data transmitted between systems.
AES-256	Advanced Encryption Standard 256-bit: a strong symmetric encryption algorithm used to secure sensitive data.
Edge Node	A local processing unit that performs real-time inference, automation logic, and temporary data buffering.
Cloud Layer	Centralized infrastructure used for analytics, dashboards, long-term storage, and model training.
Digital Twin	A real-time virtual representation of physical venue assets, occupancy, and environmental conditions.
Anomaly Detection	AI-based technique for identifying unexpected or irregular system behavior.
KPI	Key Performance Indicator: a measurable metric that evaluates operational or sustainability performance.
OTA	Over-the-Air Update: remote deployment of software updates to devices without physical access.
API	Application Programming Interface: a mechanism through which software components communicate.
Latency	The delay between sending a request and receiving a response, especially critical for real-time control.
Multi-Tenancy	System architecture enabling multiple events or venue zones to run independently within one platform instance.

Table 6- Glossary

Appendix B: Analysis Models

1. Context Diagram and Data Flow

1.1 System boundaries and external entities (Level 0)

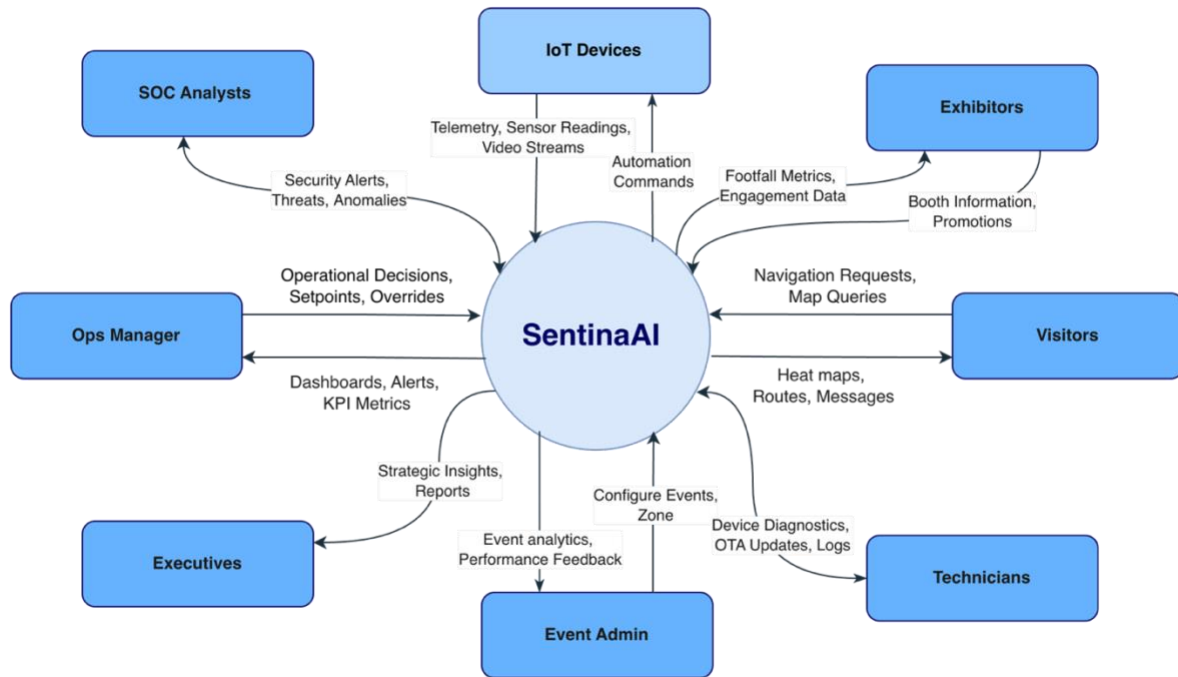


Figure 1- Context Diagram

1.2 Data flows between the system and external entities (Level 1)

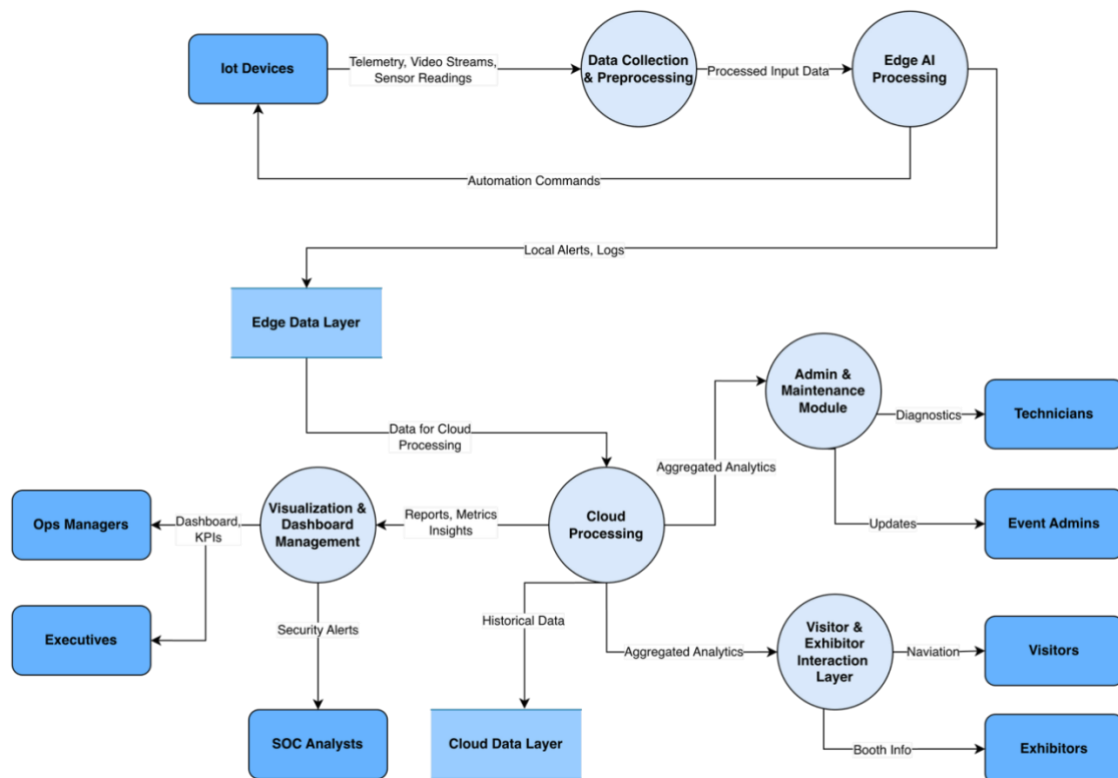


Figure 2- Data Flow Diagram

2. Use Case Descriptions

2.1 UC1: SOC Dashboard

Use case name:	SOC Dashboard
Scenario:	Detect and respond to cybersecurity incidents through the Security Operations Center (SOC) Dashboard.
Triggering event:	An anomaly or security event is detected by the Edge Node and transmitted to the Cloud Layer for analysis.
Brief description:	The SOC Dashboard allows security analysts to monitor anomalies, review system alerts, and perform attack containment actions. The system integrates data from IoT devices and edge nodes through the Cloud Layer, allowing analysts to visualize alerts and log incidents for model retraining.
Actors:	Primary Actor: SOC Analyst Supporting Actors: Edge Node, Cloud Layer, IoT Devices/Sensors, Training Dataset
Related use cases:	Botnet/DDoS Attack, Unauthorized Access
Stakeholders:	SOC Analyst, System Administrator, AI/Data Engineers, Cloud Architect/DevOps Engineer, Executive Management
Preconditions:	Edge nodes and AI models must be active, transmitting encrypted telemetry to the Cloud Layer. SOC Analyst must be logged in.
Postconditions:	The incident is contained. Audit logs and telemetry are stored in the training dataset for retraining.
Flow of activities:	Actor Step1: The SOC Analyst is alerted of an anomaly. Step2: The SOC Analyst reviews suggested containment actions computed by the AI model. Step3: The SOC Analyst confirms or triggers the containment action. Step4: The SOC Analyst monitors the containment process. Step5: The SOC Analyst closes the alert. System Step1: The Cloud Layer requests for current alerts and system status retrieved from edge telemetry. Step2: The Cloud Layer retrieves anomaly details and telemetry from IoT Devices/Sensors.

	Step3: The system presents recommended containment actions through the dashboard interface. Step4.1: The Cloud Layer issues a containment policy update to the Edge Node, which executes containment locally. Step5.1: The Edge Node confirms successful containment and synchronizes confirmation with the Cloud Layer. Step6: The Cloud Layer records the incident for the training dataset.
Exception conditions:	1. If network connectivity between Edge Node and Cloud Layer is lost, the Edge Node enforces local containment autonomously until synchronized again. 2. If the containment action fails, the SOC Analyst is notified for manual intervention.

Use Case 1- SOC Dashboard

2.2 UC2: User Portals

Use case name:	User Portals (Visitor and Exhibitor)
Scenario:	Provide real-time data visualization and analytics dashboards for visitors and exhibitors to simplify navigation and engagement insights within the venue.
Triggering event:	A registered user (visitor or exhibitor) logs into the respective portal to access live venue data or booth performance metrics.
Brief description:	The SentinaAI system includes dedicated user portals for Visitors and Exhibitors. The Visitor Portal provides live navigation, crowd heatmaps, and safety alerts using inputs from IoT sensors and the Crowd Analytics Engine. The Exhibitor Portal delivers analytics dashboards summarizing booth engagement, visitor dwell time, visit trends, and lead generation metrics powered by the Data Warehouse.
Actors:	Primary Actors: Visitor, Exhibitor Supporting Actors: Crowd Analytics Engine, IoT Sensors/Cameras, Navigation System, Data Warehouse
Related use cases:	Operations Dashboard, Sustainability Dashboard
Stakeholders:	Visitor, Exhibitor, Event Administrator, UI/UX Designer, Cloud Architect/DevOps Engineer, Software Engineer (Backend/Frontend)
Preconditions:	Analytics systems (Crowd Analytics Engine and Data Warehouse) must be operational, and IoT sensors must be actively collecting real-time data.

Postconditions:	Visitors successfully navigate the venue and receive live crowd and safety insights; Exhibitors receive accurate engagement analytics and exportable lead summaries.
Flow of activities:	Actor Visitor Portal Sequence: Step1: The Visitor logs into the Visitor Portal. Step2: The Visitor Portal requests real-time occupancy and crowd flow data from the Crowd Analytics Engine, which aggregates inputs from IoT Sensors / Cameras. Step3: The Portal displays heatmaps and crowd flow patterns Step4: The Visitor selects a destination; the Portal requests an optimized route from the Navigation System using live congestion data. Step5: The Portal provides Safety Alerts and alternate route recommendations as the visitor moves through the venue. System Step1: The system authenticates the visitor and initializes data synchronization with the Crowd Analytics Engine. Step2: The Crowd Analytics Engine retrieves live sensor data and computes occupancy density across zones. Step3: The system provides visual heatmaps and congestion overlays on the Visitor Portal interface. Step4: The Navigation System computes the optimized route based on current congestion and transmits it to the Visitor Portal. Exhibitor Portal Sequence: Step1: The Exhibitor logs into the Exhibitor Portal. Step2: The Portal retrieves booth engagement metrics, visit trends, and dwell time data from the Data Warehouse, which compiles inputs. Step3: The Portal displays engagement analytics, visit trends, and lead generation summaries. Step4: The Exhibitor exports lead data and reports for post-event follow-up. System Step1: The system authenticates the exhibitor and initializes access to their assigned booth dataset. Step2: The Data Warehouse aggregates booth analytics and crowd data to calculate engagement and dwell time.

	Step3: The system visualizes analytics in real time through the Exhibitor Dashboard. Step4: The Data Warehouse generates downloadable reports and lead summaries for export.
Exception conditions:	1. If live data from IoT Sensors/Cameras or analytics engines is unavailable, the portal displays cached or static data until synchronization resumes. 2. If navigation fails to update due to network latency, the Visitor Portal defaults to predefined routes. 3. If the Data Warehouse is temporarily unavailable, the Exhibitor Portal notifies users that analytics are being updated.

Use Case 2- User Portals

2.3 UC3: Operations Dashboard

Use case name:	Operations Dashboard
Scenario:	Regulate venue occupancy and environmental comfort through the Operations Dashboard to ensure smooth operations.
Triggering event:	Real-time occupancy or environmental data from IoT sensors is analyzed by the Edge Node, or an Operations Manager logs in to observe live venue status.
Brief description:	The Operations Dashboard enables operations managers to view live occupancy levels, detect anomalies, and initiate automated responses through the edge-based rule engine. It consolidates sensor and camera data into visual dashboards that display crowd density, temperature, and other indicators. Automated adjustments such as lighting or HVAC are imposed locally by the Edge Node, with manual override available if automation fails.
Actors:	Primary Actor: Operations Manager Supporting Actors: Edge Node, IoT Sensors/Cameras
Related use cases:	SOC Dashboard, Sustainability Dashboard
Stakeholders:	Facility Operations Manager, Maintenance Technician, System Administrator, Cloud Architect/DevOps Engineer, Executive Management
Preconditions:	Edge Nodes and IoT sensors must be operational and transmitting real-time data. The Operations Dashboard interface must have active connectivity to the edge network. The Ops Manager should be logged in.
Postconditions:	Venue occupancy and comfort conditions are stabilized; alerts are cleared, and logs are updated for performance analysis.

Flow of activities:	Actor Step1: The Operations Manager logs into the Operations Dashboard. Step2: The Operations Manager views live occupancy and comfort metrics. Step3: The Operations Manager reviews alerts or notifications shown on the dashboard. Step4: The Operations Manager triggers an automated response when an anomaly is detected. Step5: If automation fails, the Operations Manager performs a manual override. System Step1.1: The Dashboard retrieves real-time data from IoT Sensors / Cameras. Step2.1: The Edge Node continuously analyzes the data for anomalies such as overcrowding or temperature deviation. Step3.1: When thresholds are exceeded, the Edge Node sends an alert to the Operations Dashboard. Step4.1: Upon user confirmation, the Edge Node executes the automated action locally. Step5.1: The Dashboard confirms successful execution and updates the visual system status in real time.
Exception conditions:	1. If automation fails or network latency prevents command execution, the Operations Manager manually overrides the system through the Dashboard interface. 2. If communication between the Edge Node and IoT Sensors is interrupted, cached rules continue operating locally until synchronization resumes.

Use Case 3- Operations Dashboard

2.4 UC4: Sustainability Dashboard

Use case name:	Sustainability Dashboard
Scenario:	Analyze and track energy consumption, HVAC performance, and environmental metrics to improve sustainability.
Triggering event:	The Sustainability Manager logs into the dashboard to view energy analytics or scheduled sustainability reports are generated automatically.
Brief description:	The Sustainability Dashboard visualizes current and historical energy data collected from IoT-based energy monitors, HVAC systems, and environmental sensors. It provides analytics on consumption trends, carbon emissions, and predictive insights on HVAC demand to support sustainability planning.
Actors:	Primary Actor: Sustainability Manager Supporting Actors: Cloud Layer, Energy Monitoring Devices, Environmental Sensors, Data Warehouse
Related use cases:	Operations Dashboard, SOC Dashboard
Stakeholders:	Facility Operations Manager, Executive Management, AI/Data Engineers, Cloud Architect/DevOps Engineer,
Preconditions:	Energy monitoring devices, HVAC data streams, and environmental sensors must be operational and synchronized with the cloud analytics system. User should be logged in.
Postconditions:	Updated sustainability reports and energy policy insights are stored; system optimization recommendations are applied where necessary.
Flow of activities:	Actor Step1: The Sustainability Manager opens the Sustainability Dashboard Step2: The Manager views current and historical energy usage data. Step3: The Manager reviews energy savings, HVAC forecasts, and environment metrics. Step4: The Manager generates a sustainability report for internal review or compliance submission. System Step1.1: The Cloud Layer aggregates energy data and HVAC telemetry from monitoring devices. Step2.1: The Data Warehouse processes historical datasets and computes energy efficiency indicators.

	Step3.1: The AI engine forecasts HVAC demand and carbon emission trends. Step4.1: The Dashboard visualizes metrics and stores generated reports for access or auditing.
Exception conditions:	1. If data synchronization is delayed, the dashboard displays the most recent available readings with timestamps. 2. If any sensor or energy feed is offline, the dashboard flags the missing data source and uses cached values until the connection is restored.

Use Case 4- Sustainability Dashboard

2.5 UC5: Botnet/DDoS Attacks

Use case name:	Botnet/DDoS Attacks
Scenario:	Detect and contain a DDoS or Botnet attack targeting networked IoT devices, assuring uninterrupted operation of the venue's systems.
Triggering event:	Unusual network traffic or device behaviour is detected by the Edge Node's monitoring engine, indicating a potential DDoS or botnet attack.
Brief description:	When a botnet attack attempts to overwhelm connected IoT devices (such as CCTV systems), the Edge Node identifies the abnormal traffic using anomaly detection models. The rule engine flags the activity as a DDoS attempt and isolates the compromised nodes locally. An alert is automatically sent to the SOC Dashboard for review, where the analyst validates containment and logs the event.
Actors:	Primary Actor: SOC Analyst Supporting Actors: Edge Node, IoT Devices / Sensors, Cloud Layer
Related use cases:	SOC Dashboard, Unauthorized Access Detection
Stakeholders:	SOC Analyst, System Administrator, Cloud Architect/DevOps Engineer, AI/Data Engineers
Preconditions:	Edge Nodes and AI inference models must be operational and continuously monitoring device-level network traffic.

Postconditions:	Affected devices are quarantined; network traffic stabilizes; incident logs are updated in the SOC Dashboard for audit and future training data.
Flow of activities:	Actor Step1: The SOC Analyst monitors real-time alerts on the SOC Dashboard. Step2: Upon notification, the Analyst reviews anomaly details and containment status. Step3: The SOC Analyst confirms containment and logs the incident. System Step1.1: The Edge Node detects abnormal traffic volume or entropy variance across IoT devices. Step1.2: The Rule Engine classifies the pattern as a potential DDoS attempt. Step1.3: The Edge Node quarantines the affected IoT devices locally to prevent lateral spread. Step1.4: An alert is sent to the SOC Dashboard through the Cloud Layer for review. Step2.1: The Cloud Layer synchronizes logs and updates the incident record once containment is confirmed.
Exception conditions:	1. If the anomaly classification model cannot confidently determine the attack, the alert is escalated for manual verification by the SOC Analyst. 2. If network connectivity to the Cloud Layer is interrupted, the Edge Node enforces local containment autonomously until synchronization resumes.

Use Case 5- Botnet/DDoS Attacks

2.6 UC6: Unauthorized Access

Use case name:	Unauthorized Access
Scenario:	Detect and respond to unauthorized login attempts, automatically locking affected accounts and preventing network compromise.
Triggering event:	Multiple failed login attempts or suspicious credential patterns are detected across monitored devices.

Brief description:	The system continuously monitors authentication logs to detect repeated failed attempts, credential reuse, or brute-force activity. When such behaviour is detected, the affected account is locked, and the originating IP address is blacklisted. The incident is reported to the SOC Dashboard, where the SOC Analyst reviews and confirms the event. After verification, legitimate users can have their accounts securely restored.
Actors:	Primary Actor: SOC Analyst Supporting Actors: Edge Node, Cloud Layer, Authentication Service
Related use cases:	SOC Dashboard, Botnet/DDoS Detection
Stakeholders:	SOC Team, IT Security, System Administrators
Preconditions:	Authentication services and access control policies must be active. Edge Nodes must be monitoring login metadata in real time.
Postconditions:	Unauthorized access is prevented, blacklisted IPs and account status are updated, and the incident is recorded for audit and model retraining.
Flow of activities:	Actor Step1: The SOC Analyst monitors the SOC Dashboard for security alerts. Step2: Upon receiving an unauthorized access alert, the Analyst reviews the affected account and login logs. Step3: The SOC Analyst verifies the action taken and confirms resolution. Step4: If verified as false positive, the SOC Analyst restores access manually through the Dashboard. System Step1.1: The Authentication Service logs all login attempts across connected systems. Step1.2: The Edge Node aggregates and analyzes login metadata for suspicious activity or brute-force patterns. Step1.3: The AI model flags abnormal login behaviour and triggers an automatic lockout of the affected account.

	Step1.4 The system blacklists the source IP and sends an alert to the SOC Dashboard via the Cloud Layer. Step1.5 The Cloud Layer synchronizes updated credentials, IP blocks, and incident details for centralized reporting.
Exception conditions:	1. If a legitimate user is locked out due to a false positive, the SOC Analyst can manually revoke the lockout and restore access via the SOC Dashboard. 2. If network latency delays alert synchronization, the Edge Node maintains the lockout policy locally until the Cloud Layer reconnects.

Use Case 6- Unauthorized Access

3. Use Case Diagrams

3.1 UC1: SOC Dashboard

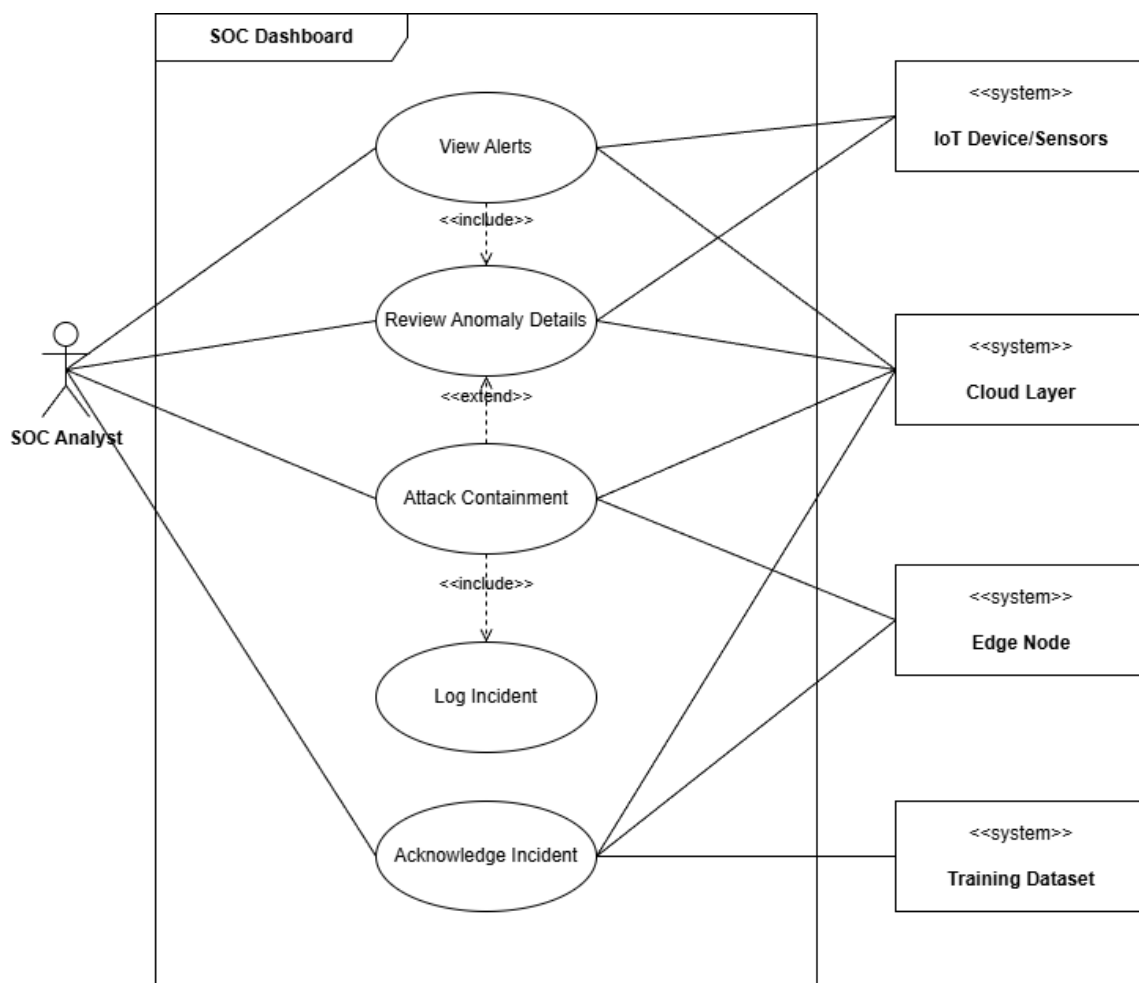


Figure 3- SOC Dashboard UCD

3.2 UC2: User Portals

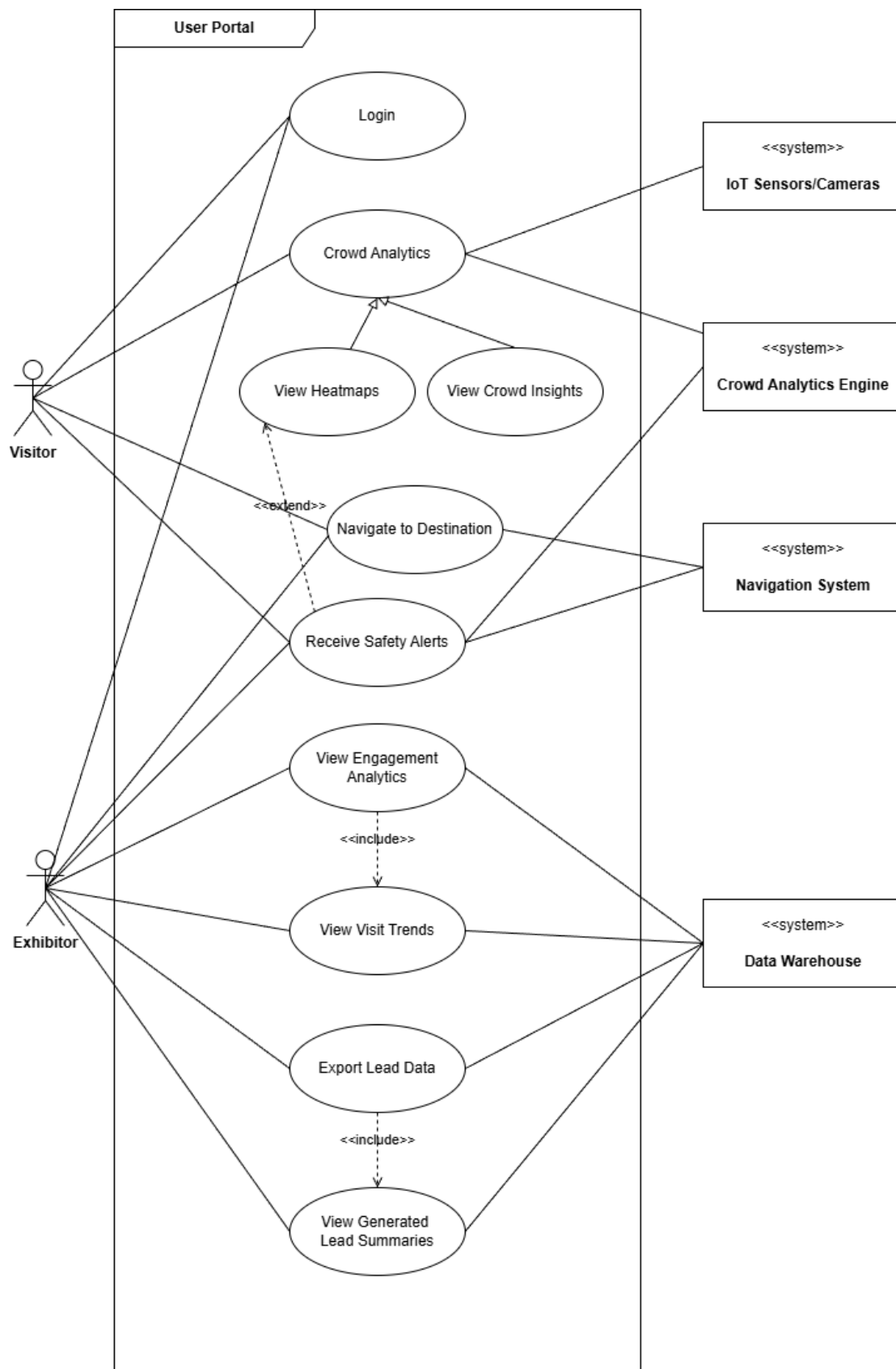


Figure 4- User Portal UCD

3.3 UC3: Operations Dashboard

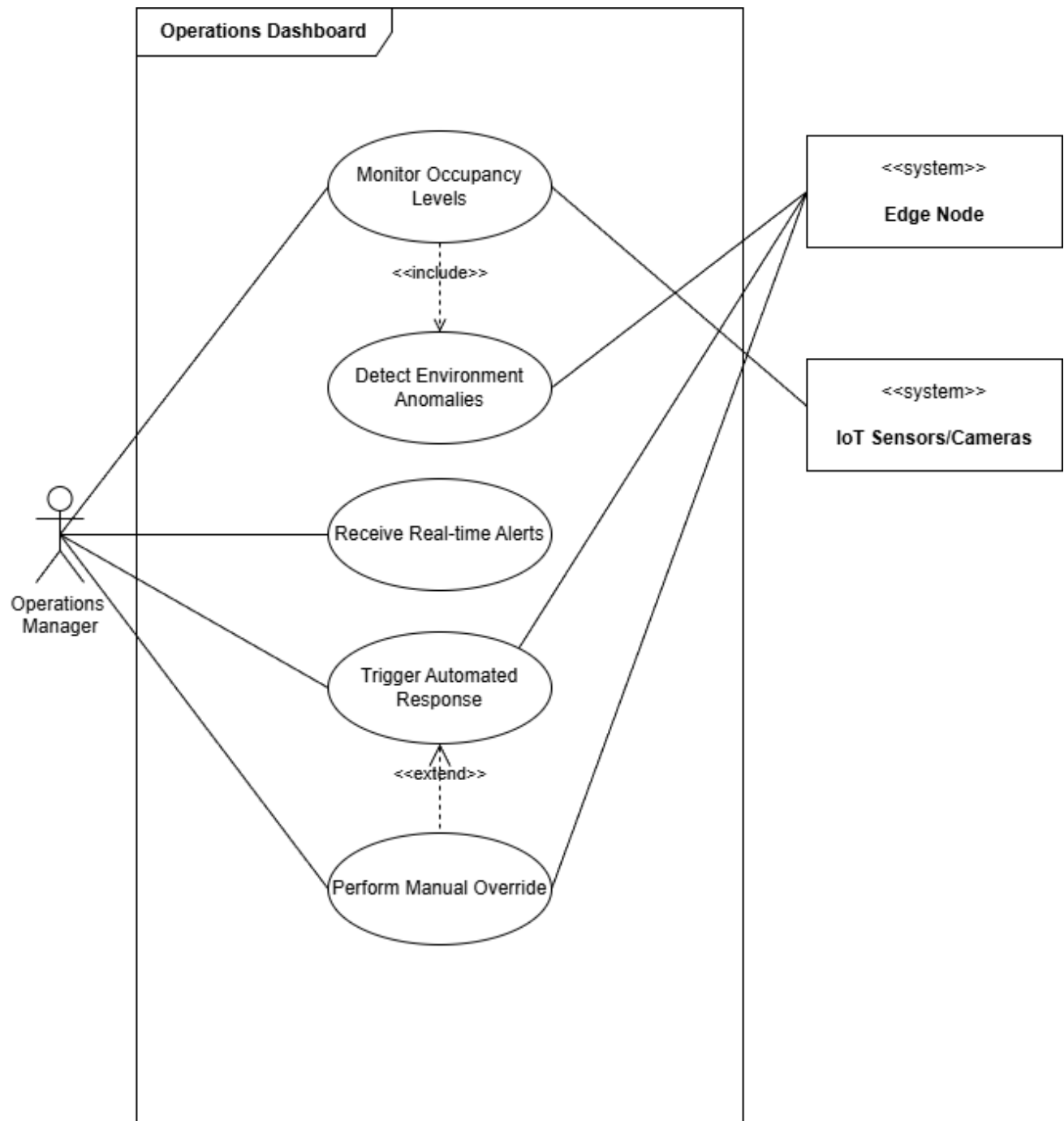


Figure 5- Operations Dashboard UCD

3.4 UC4: Sustainability Dashboard

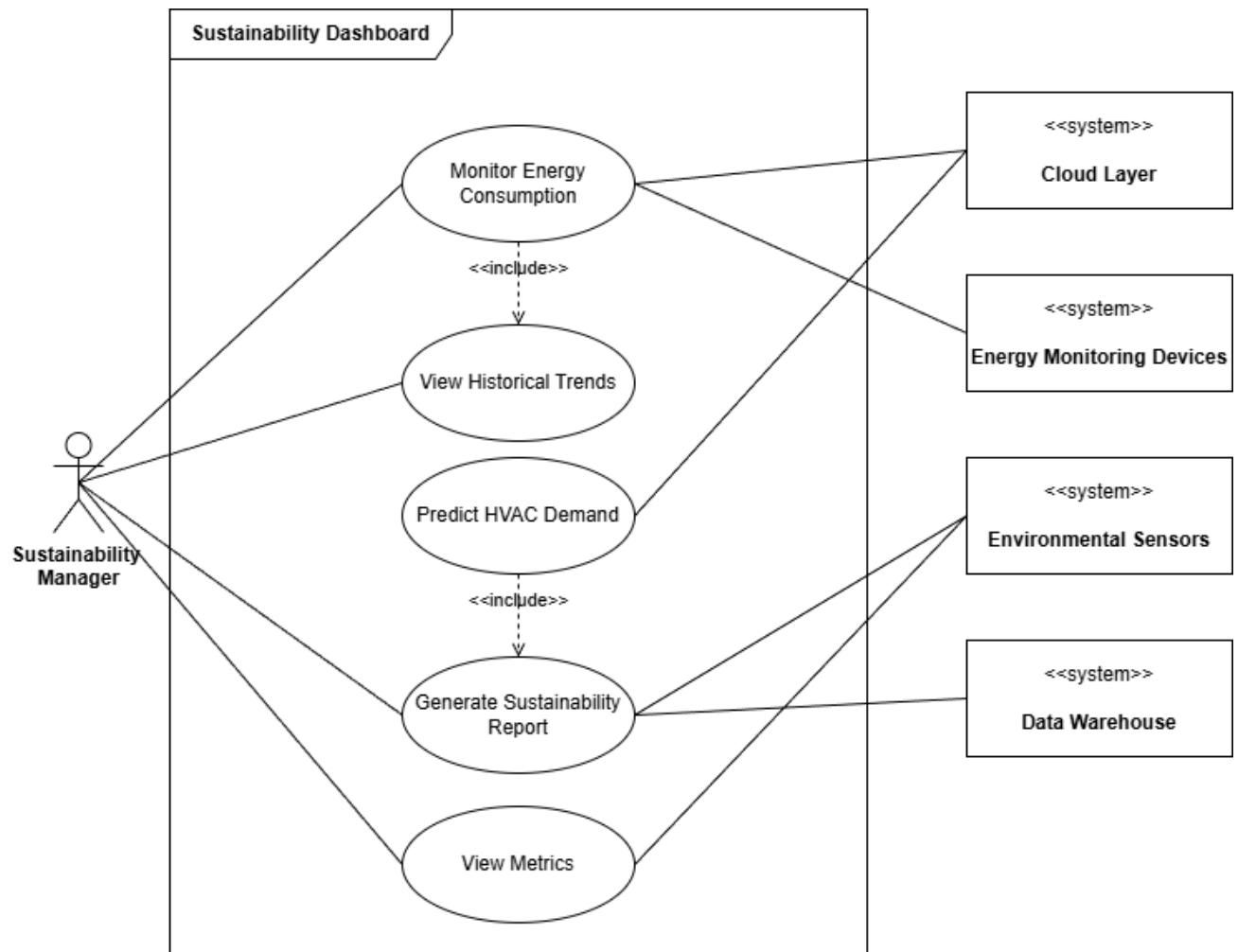


Figure 6- Sustainability Dashboard UCD

3.5 UC5: Botnet/DDoS Attacks

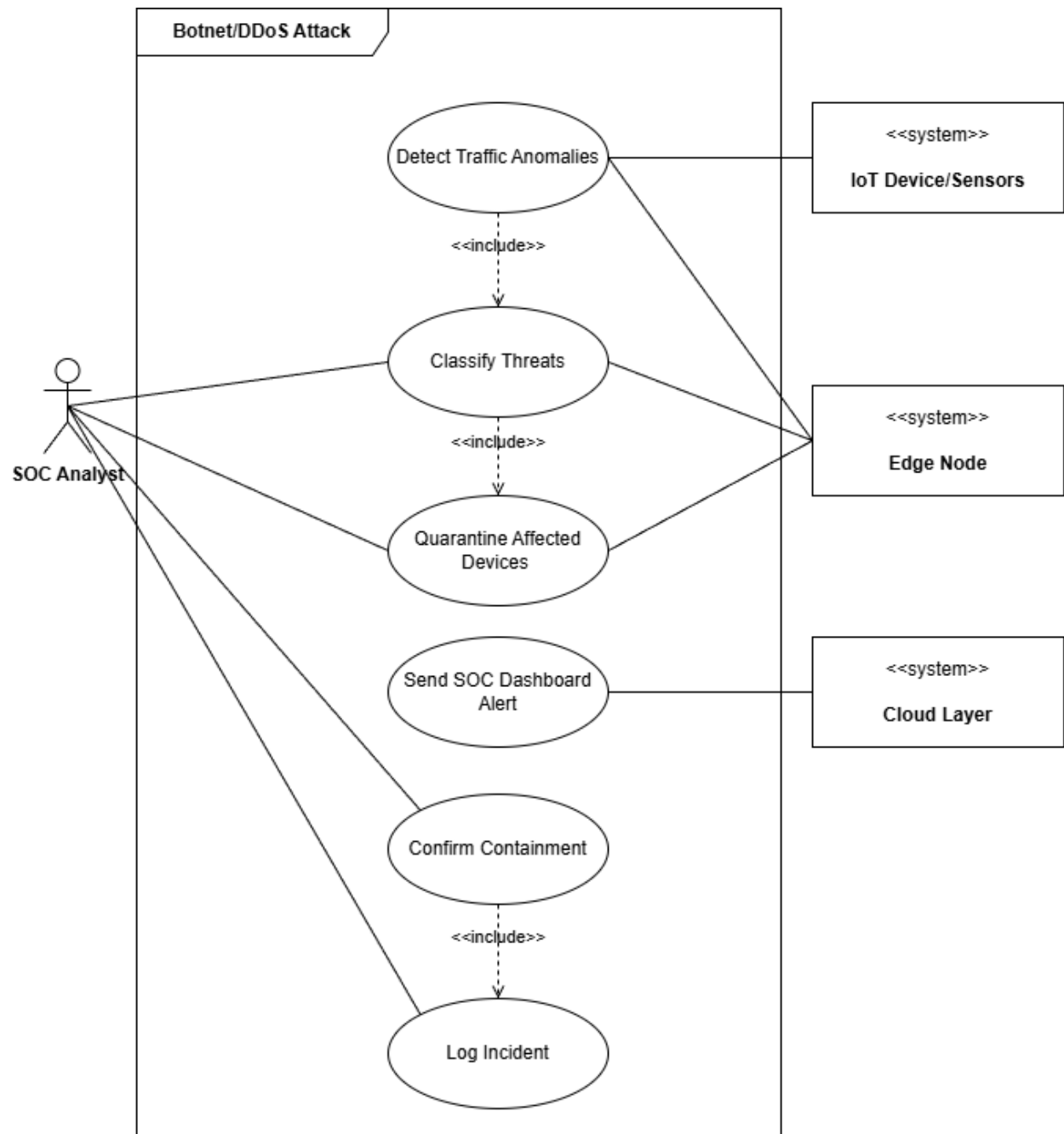


Figure 7- Botnet/DDoS Attack UCD

3.6 UC6: Unauthorized Access

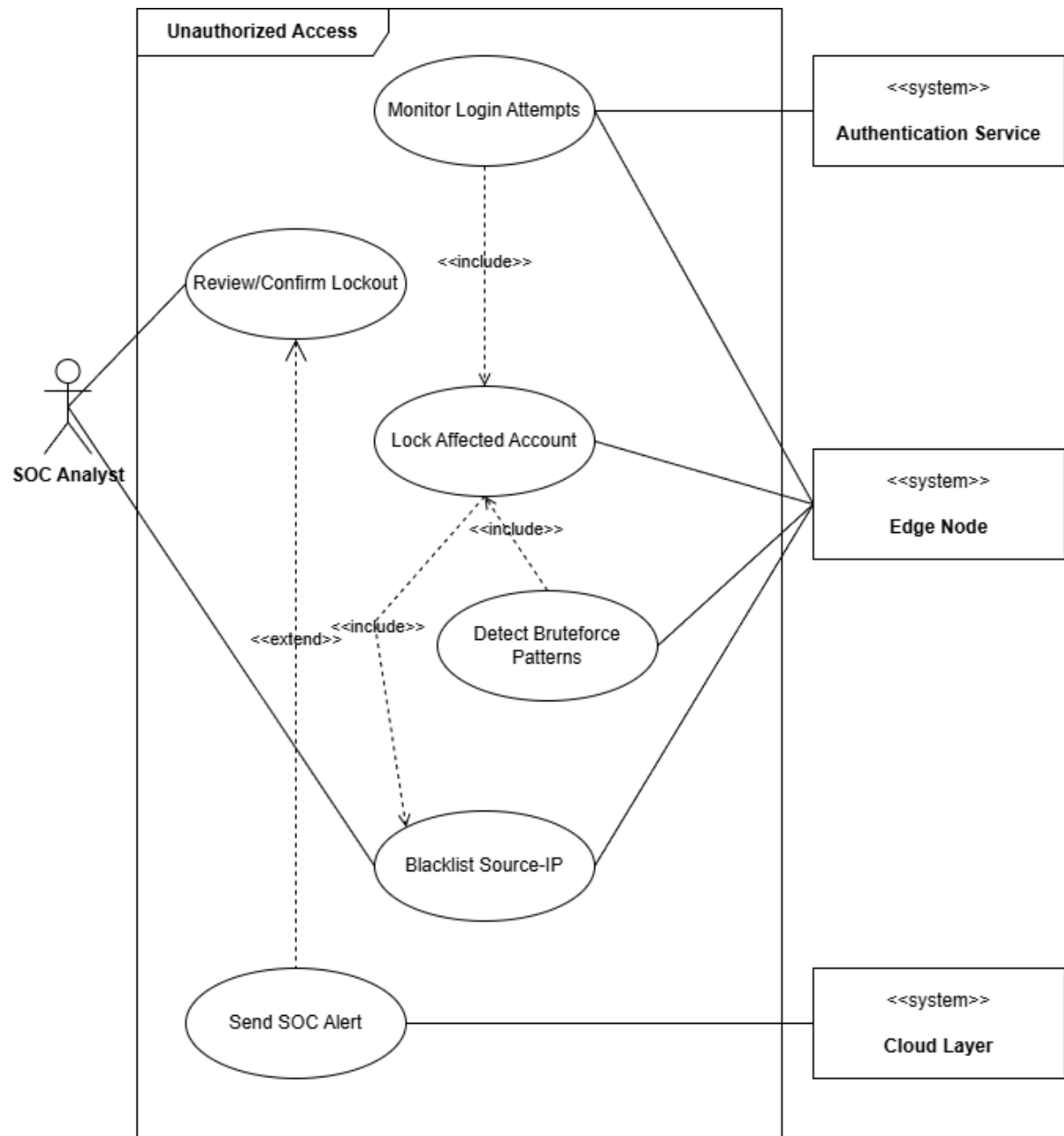


Figure 8- Unauthorized Access UCD

4. Activity Diagrams

4.1 UC1: SOC Dashboard

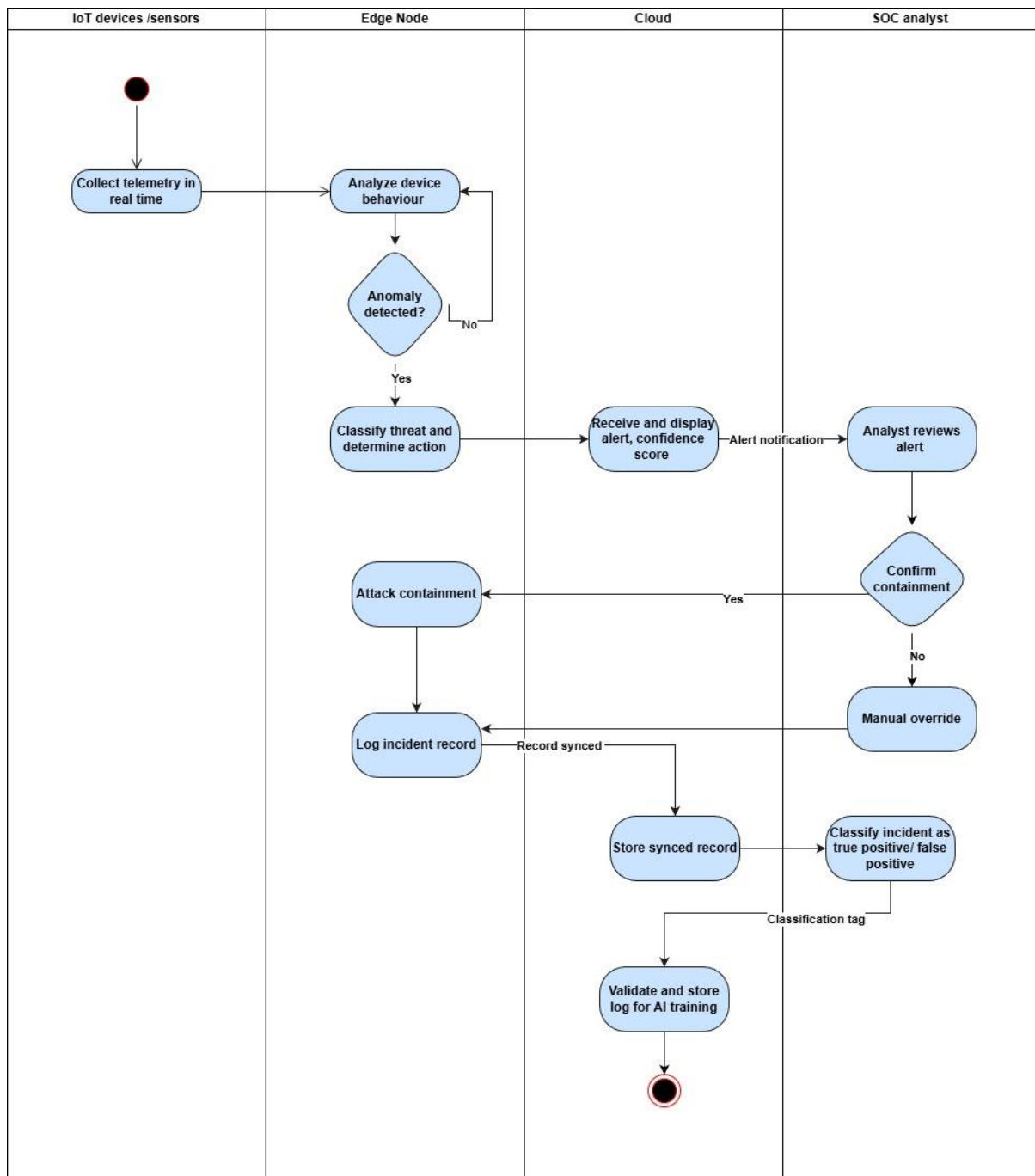


Figure 9- SOC Dashboard Activity Diagram

4.2 UC2: User Portals

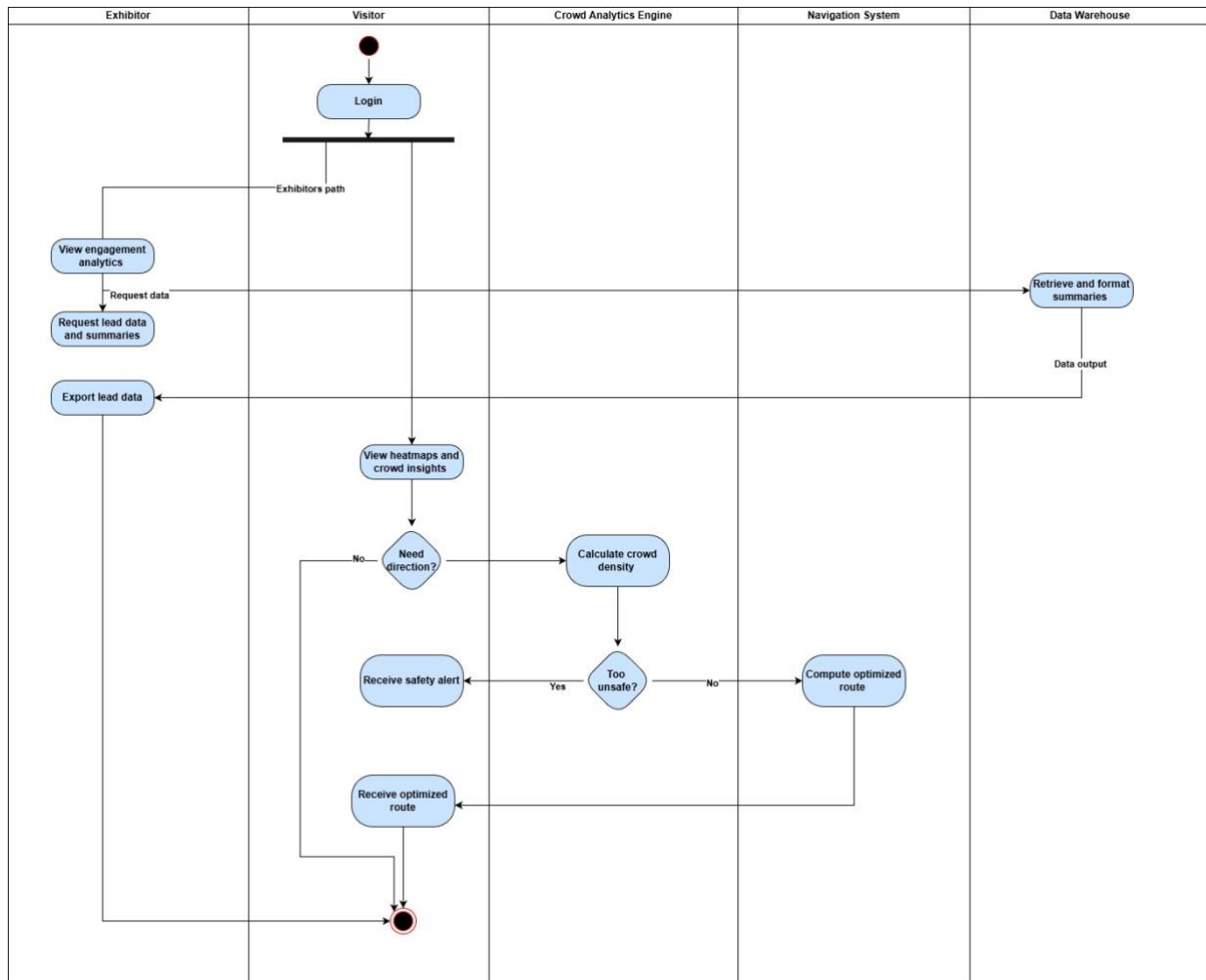


Figure 10- User Portal Activity Diagram

4.3 UC3: Operations Dashboard

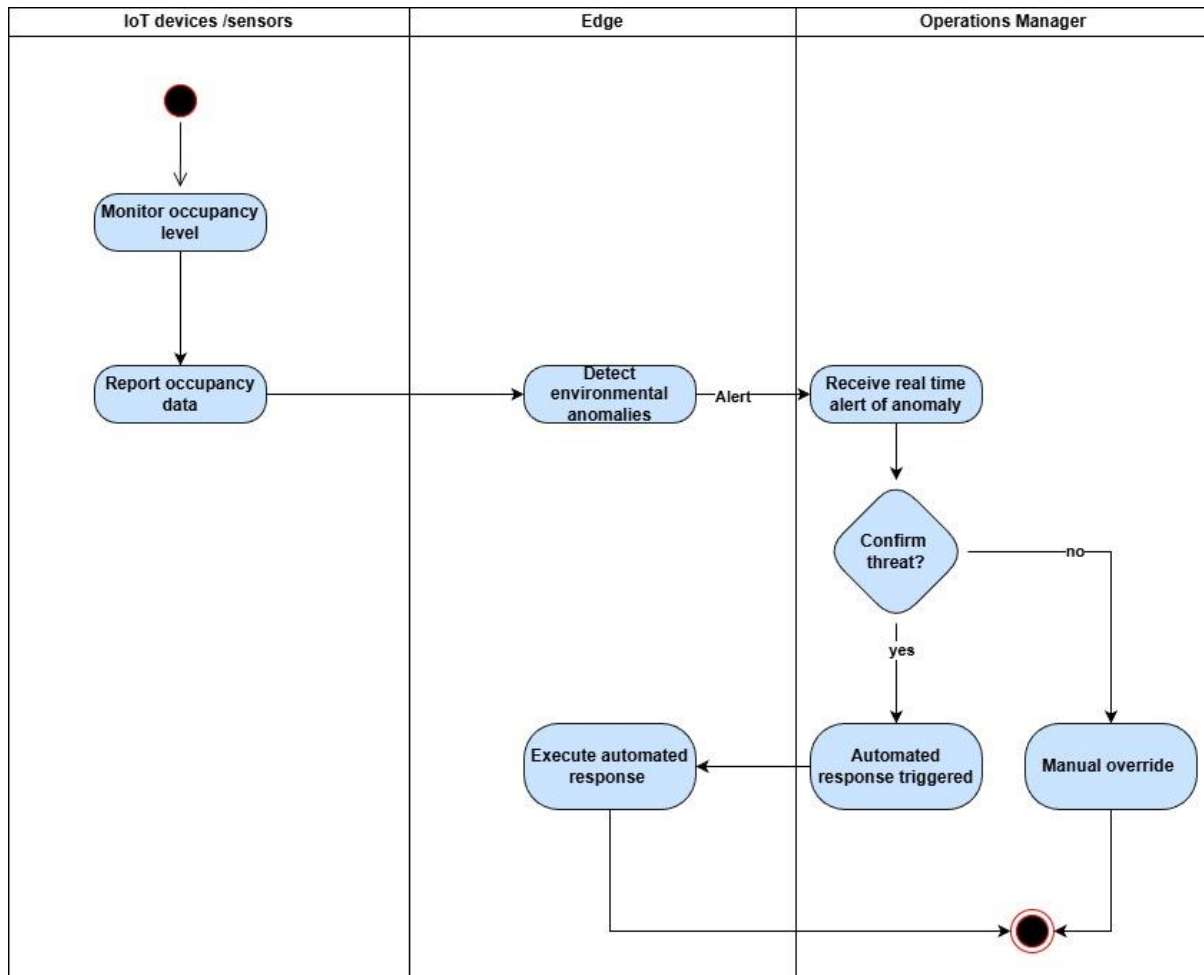


Figure 11- Operations Dashboard Activity Diagram

4.4 UC4: Sustainability Dashboard

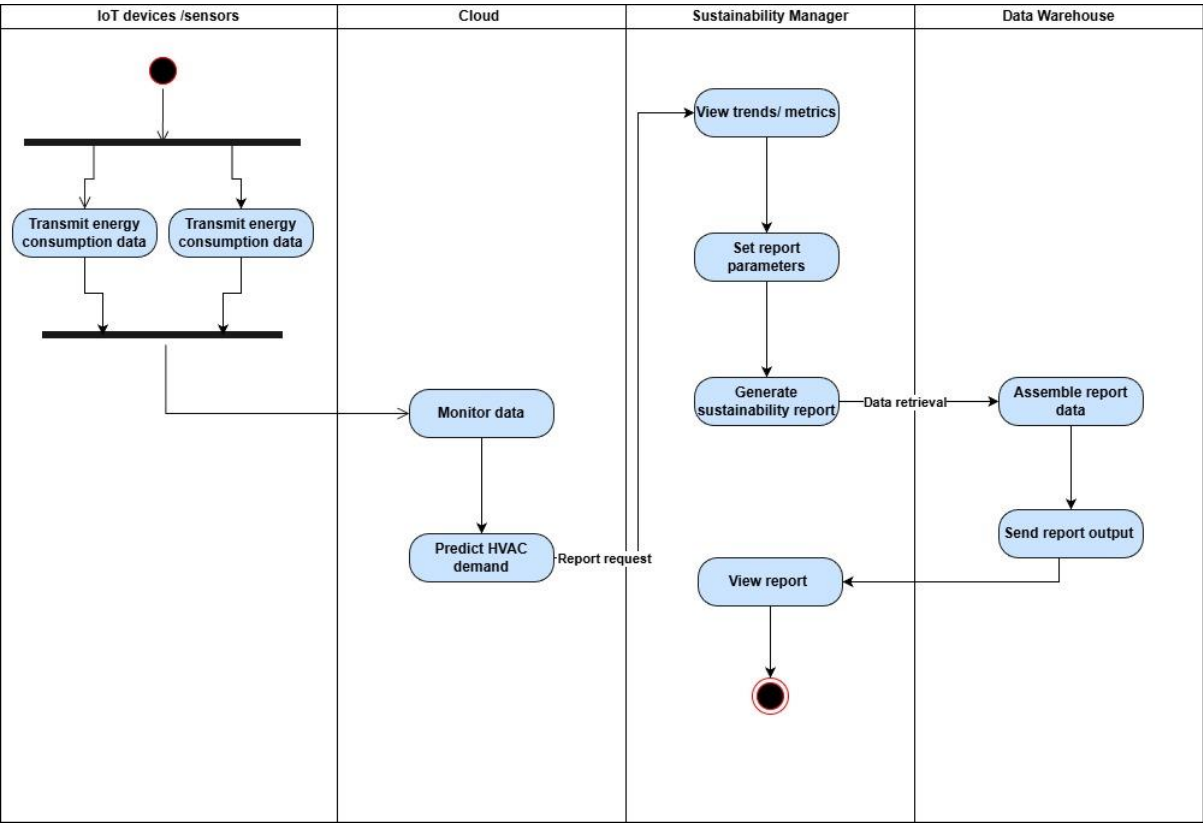


Figure 12- Sustainability Dashboard Diagram

4.5 UC5: Botnet/DDoS Attacks

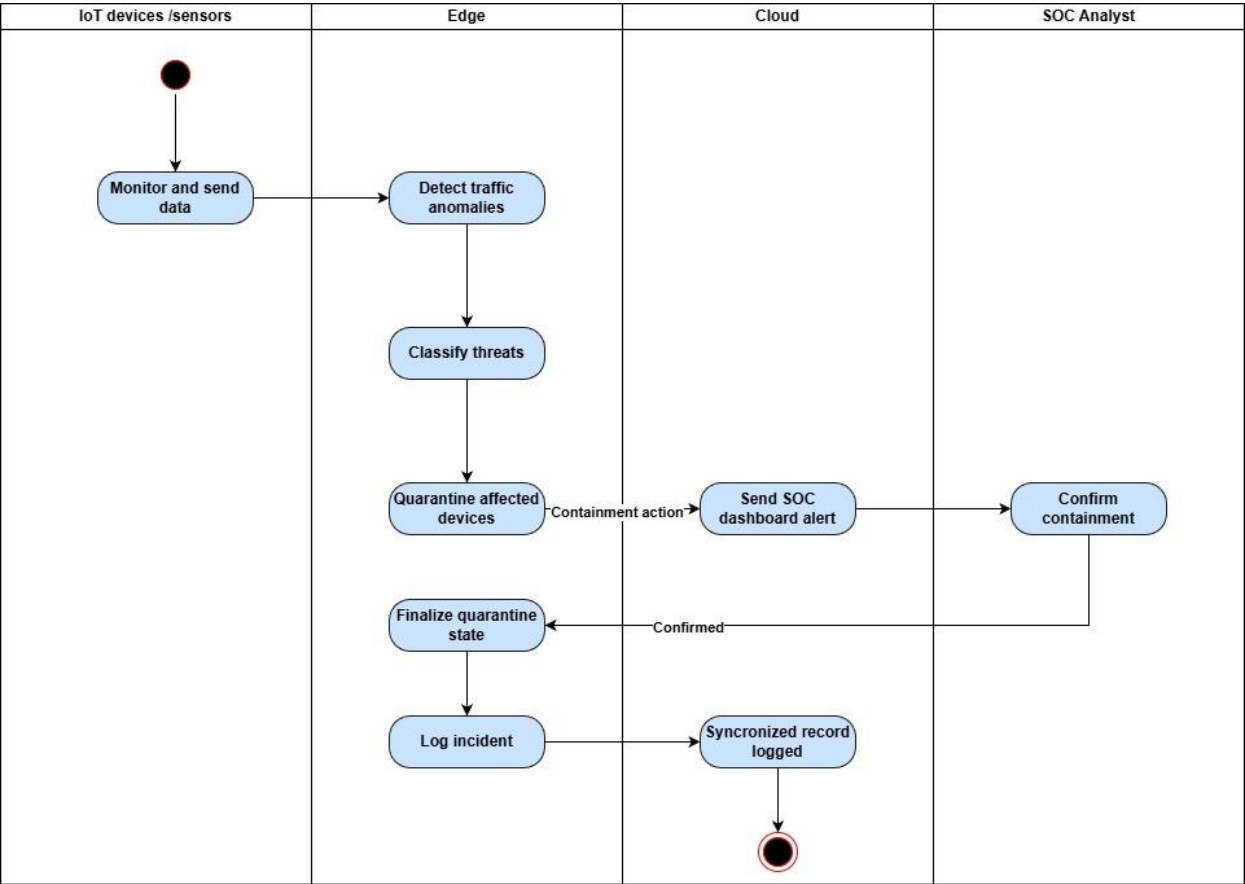


Figure 13- Botnet/DDoS Attacks

4.6 UC6: Unauthorized Access

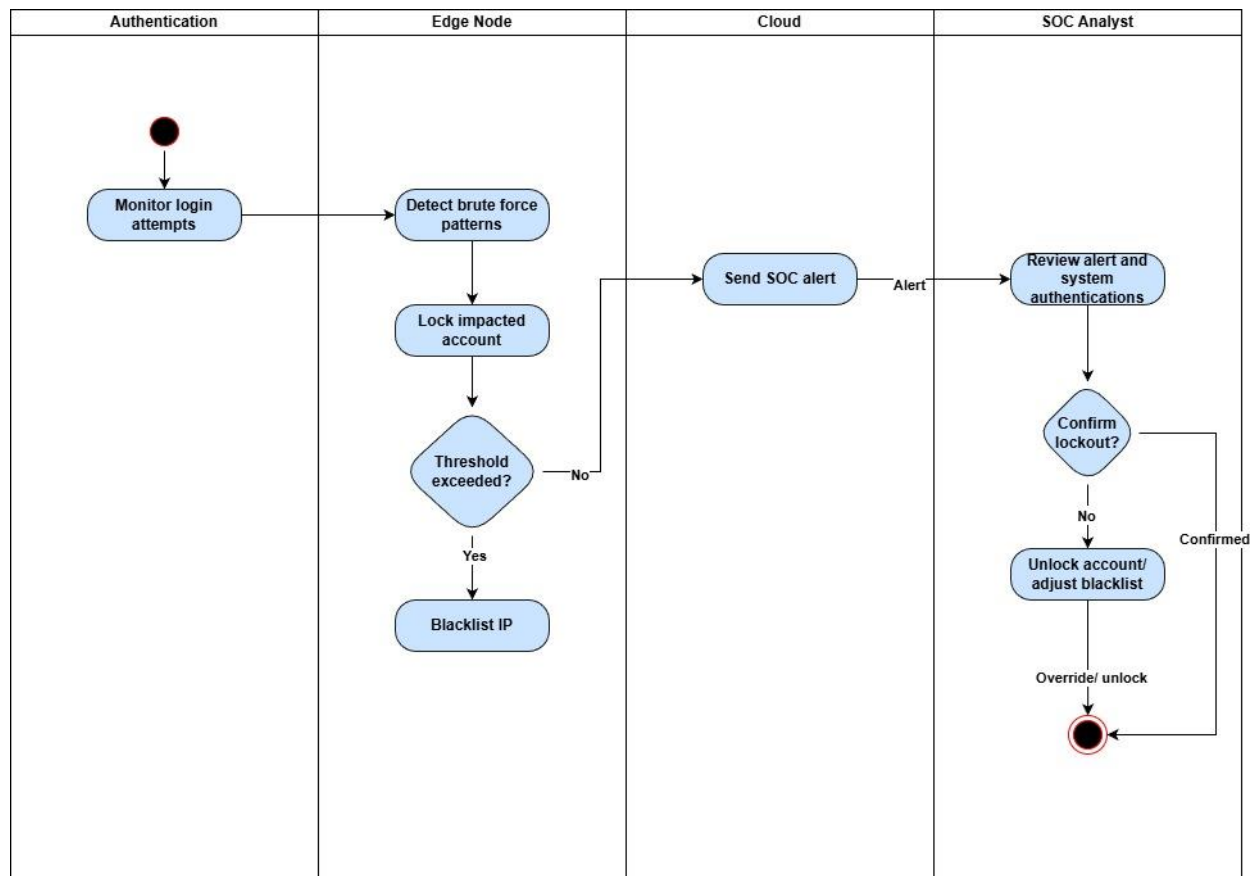


Figure 14- Unauthorized Access

5. Sequence Diagrams

5.1. UC1: SOC Dashboard

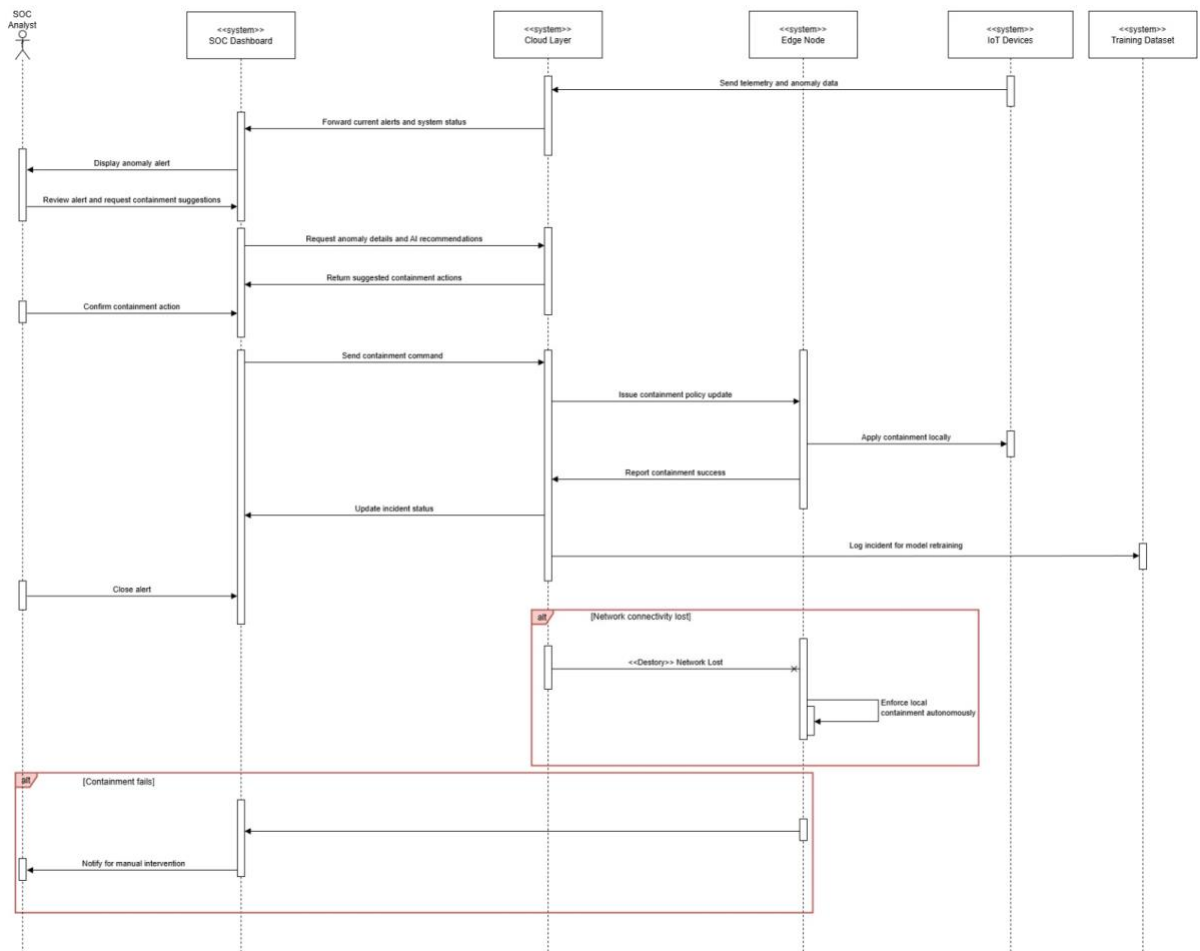


Figure 15- SOC Dashboard Sequence Diagram

5.2. UC2: User Portals (Visitor)

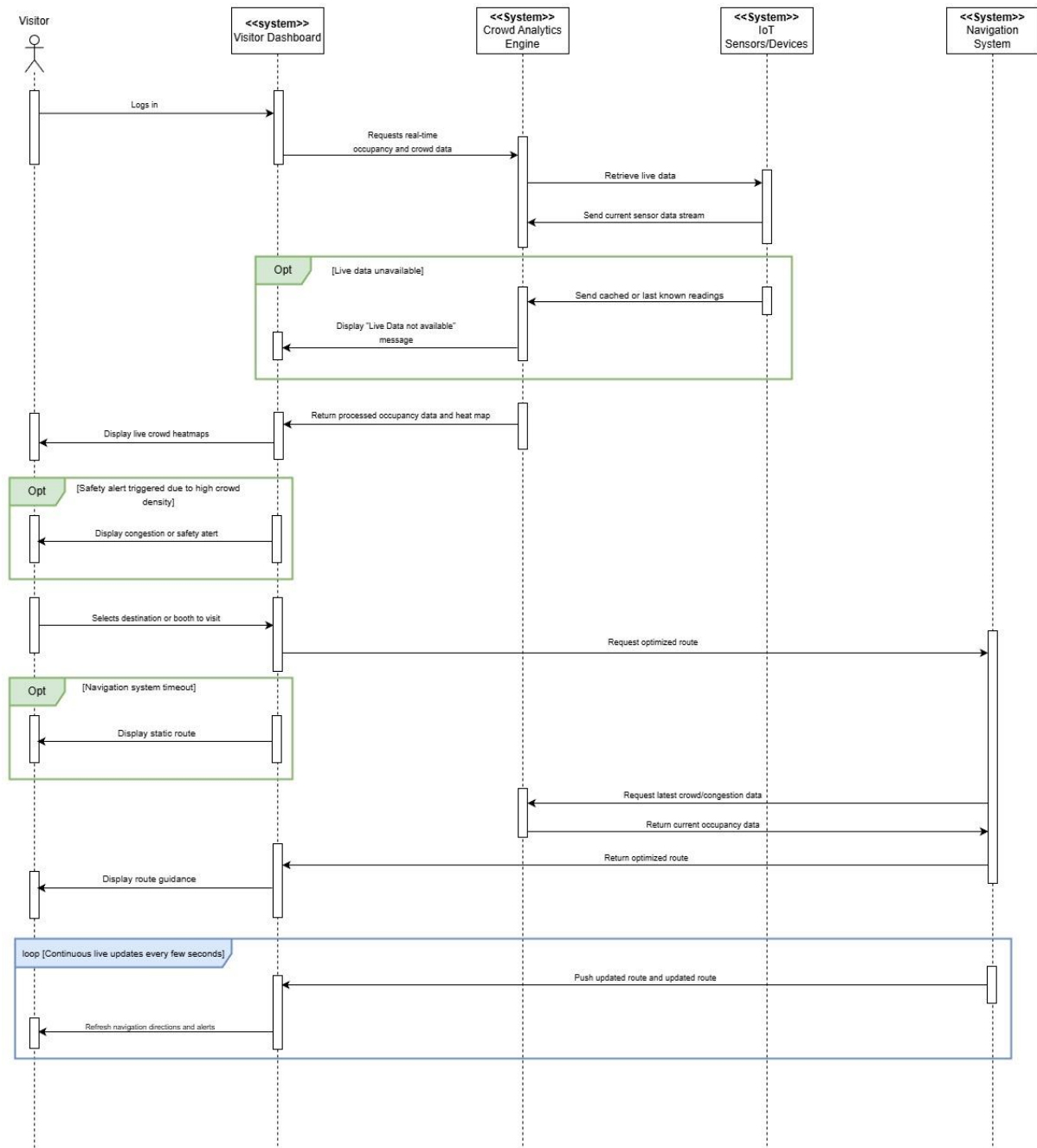


Figure 16- Visitor Portal Sequence Diagram

5.3. UC2: User Portals (Exhibitor)

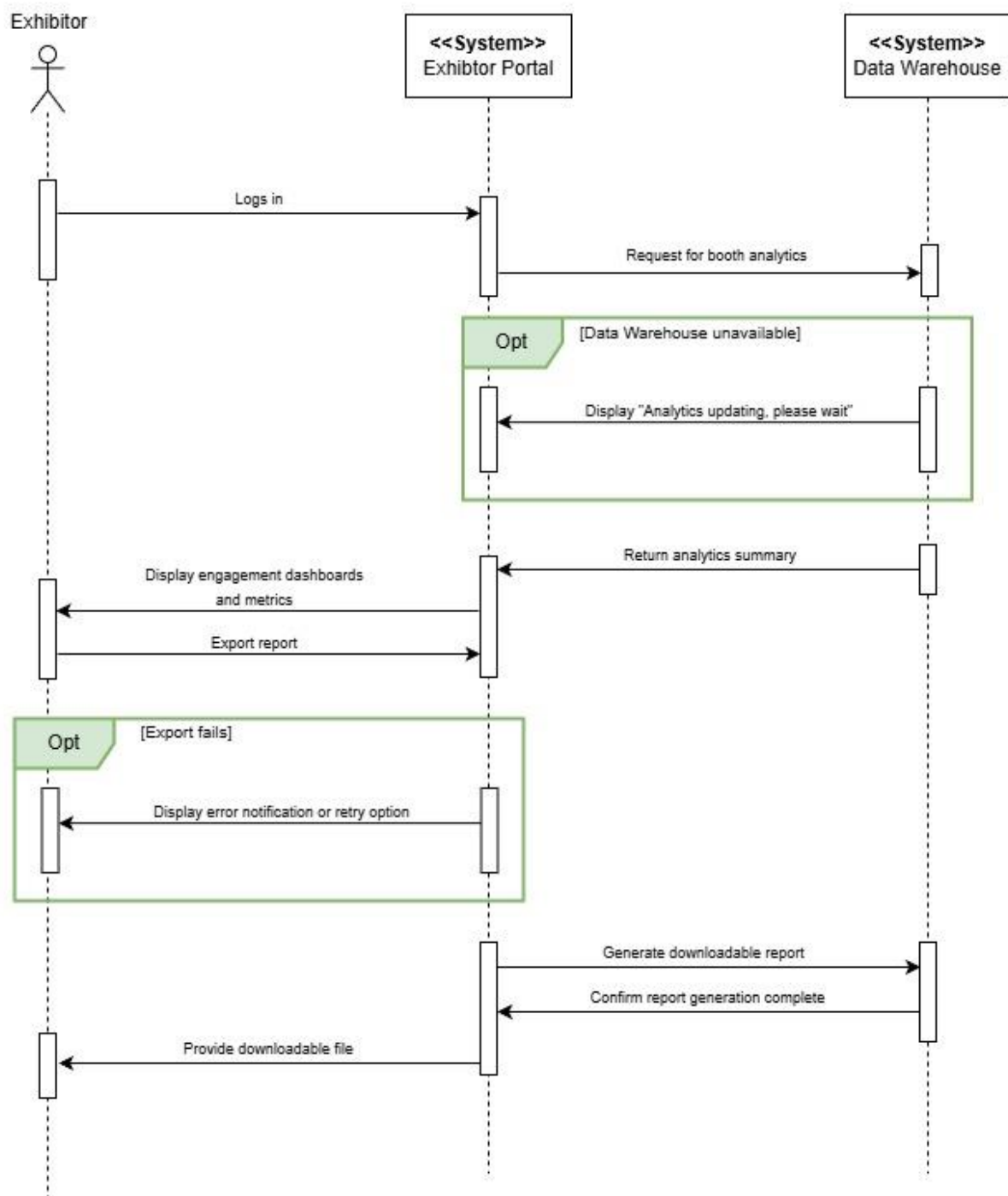


Figure 17- Exhibitor Portal Sequence Diagram

5.4. UC3: Operations Dashboard

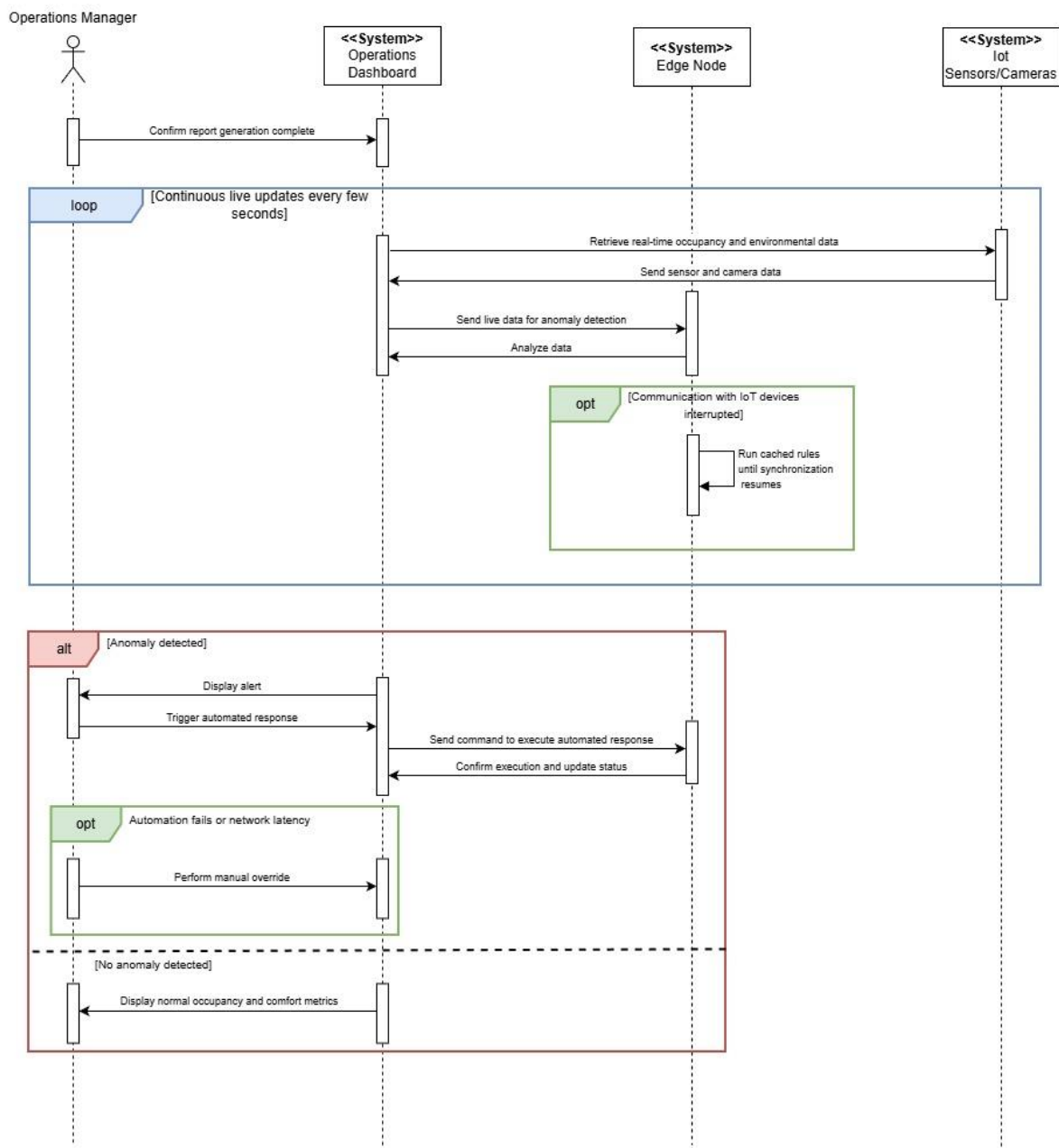


Figure 18- Operations Dashboard Sequence Diagram

5.5. Sustainability Dashboard

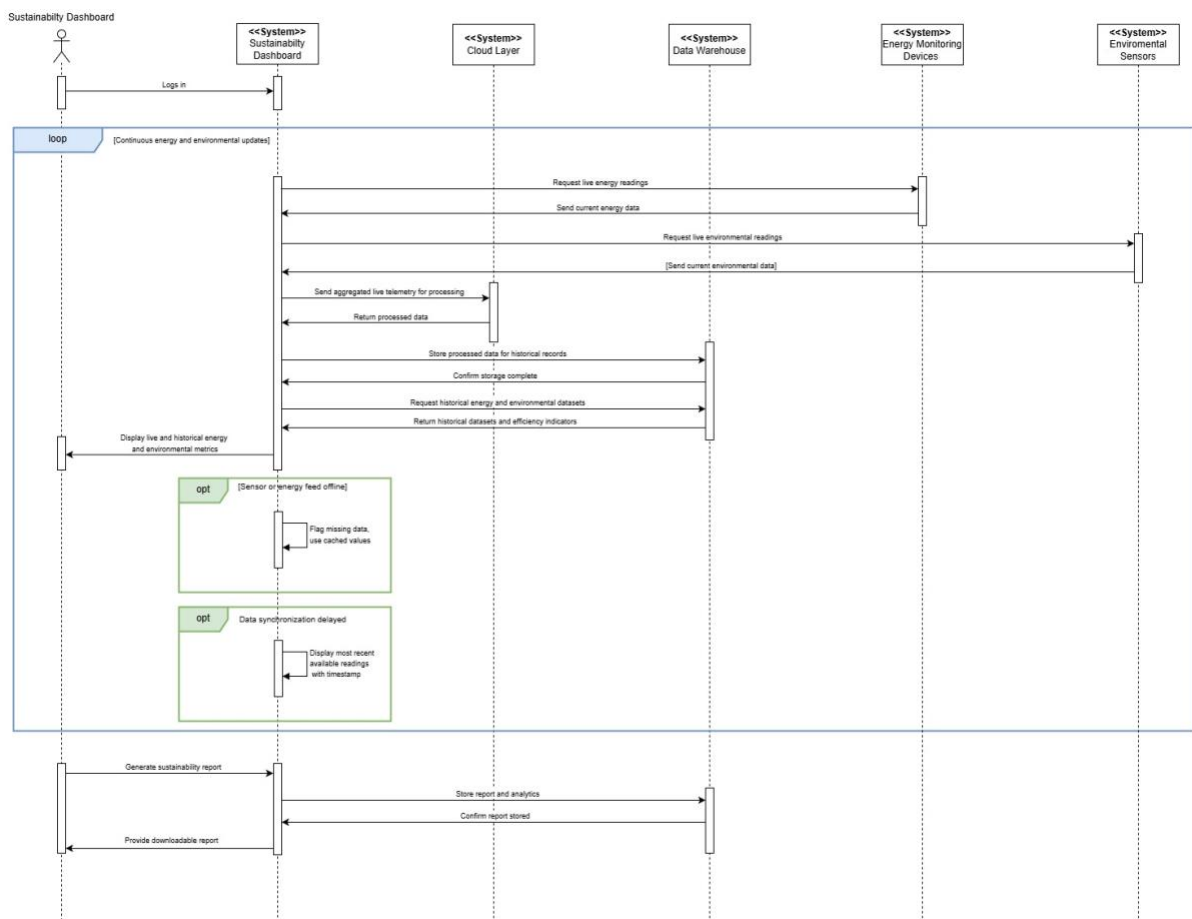


Figure 19- Sustainability Dashboard Sequence Diagram

5.6. Botnet/DDoS Attacks

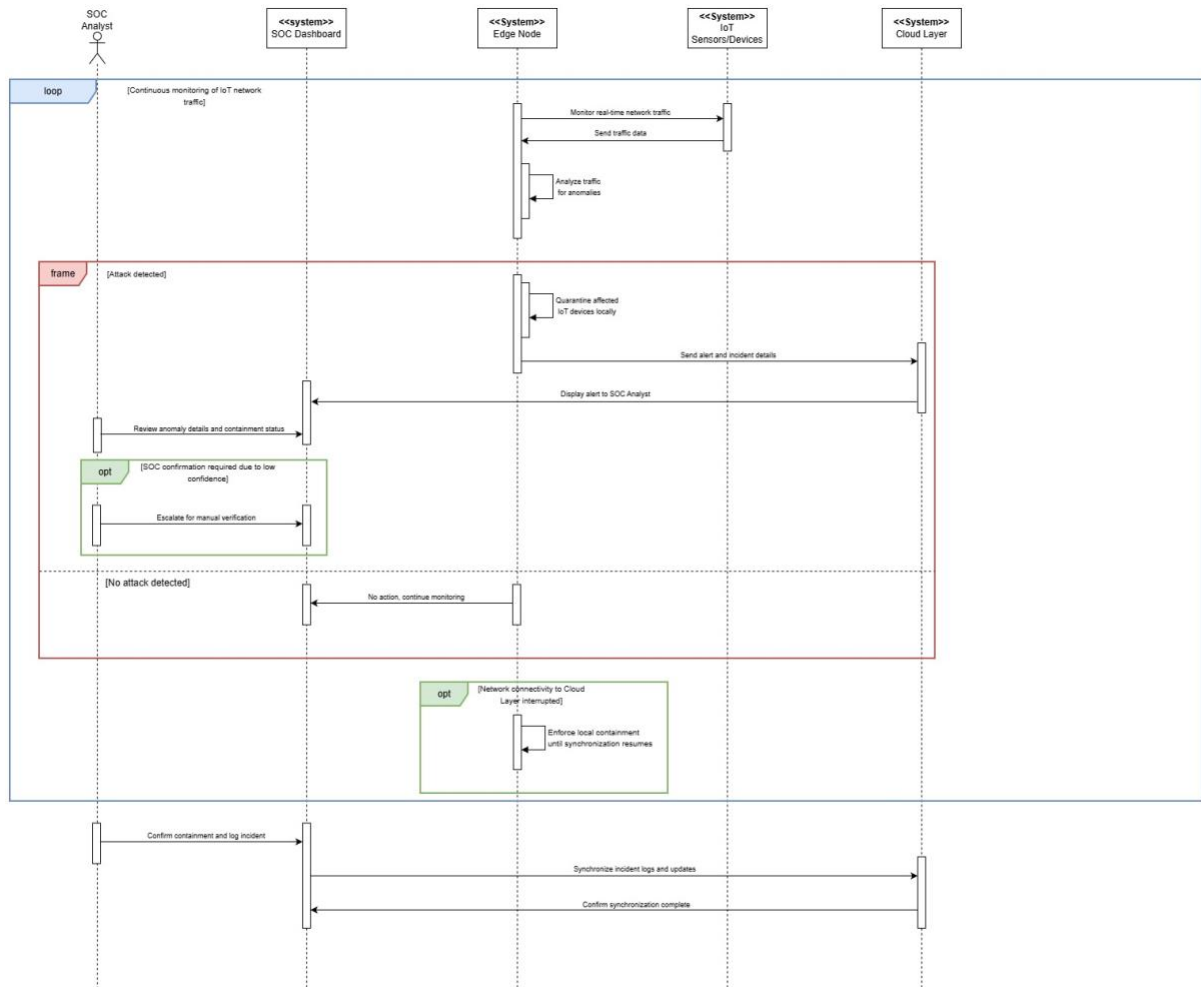


Figure 20- Botnet/DDoS Sequence Diagram

5.7. Unauthorized Access

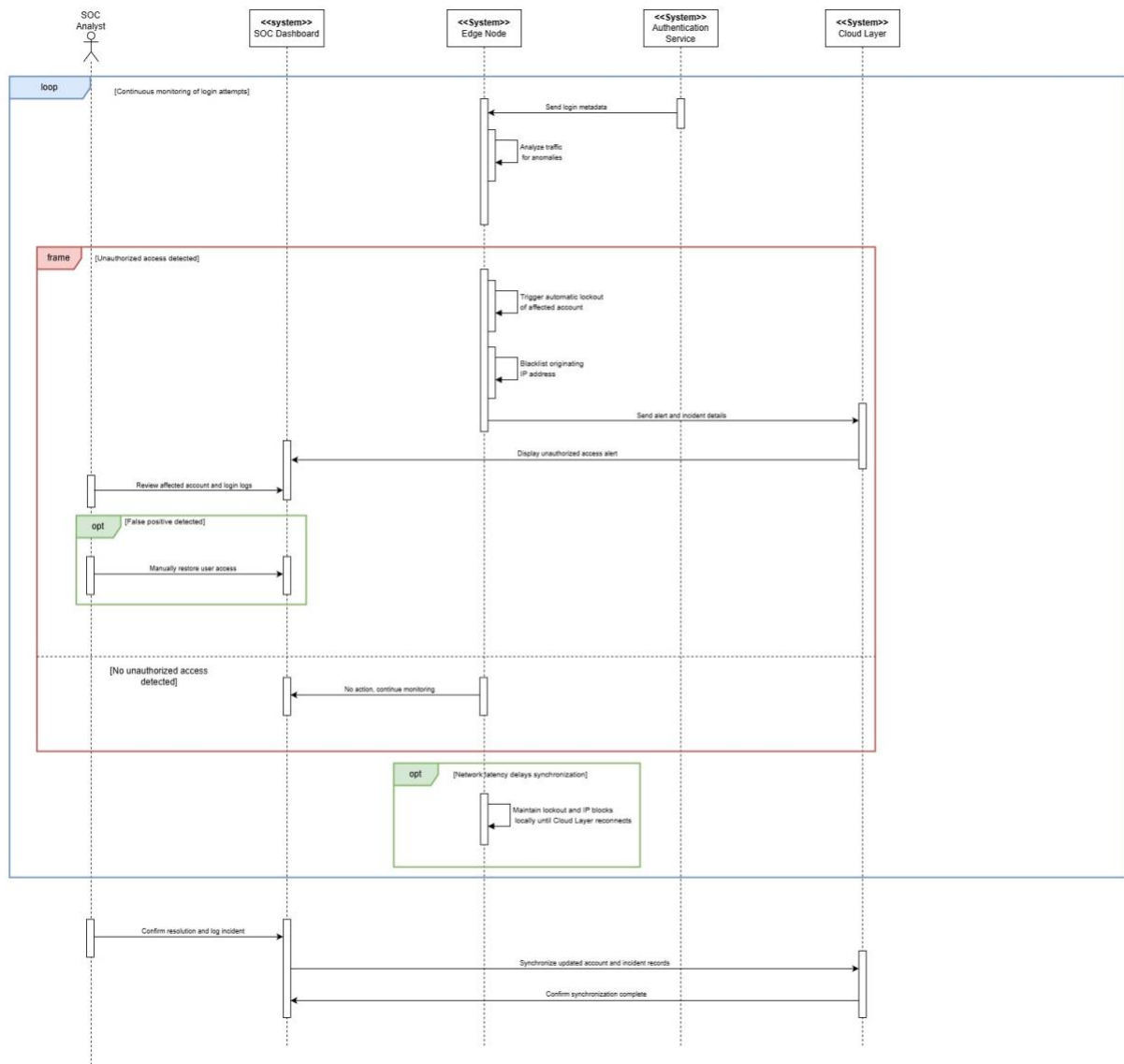


Figure 21- Unauthorized Access Sequence Diagram

Appendix C: Issues List

The following items represent open requirements issues that are not yet fully resolved.. This list will be updated throughout the project lifecycle.

1. **Privacy:** Collecting and analyzing data live from thousands of IoT devices especially cameras and environmental sensors compromises privacy. We would need to implement privacy-by-design architectures that complies with UAE's PDPL.
2. **Realism of Datasets available:** One of the most crucial gaps to fill moving forward is the lack of large realistic IoT security and facility management datasets. The public datasets available (i.e., BoT-IoT, TON-IoT) are synthetically generated or data collected in lab settings, meaning they are not as realistic as needed. Therefore, model training might not be extensive enough to consider the scenario variability needed in such high-density, fluid IoT environments such as convention centers.
3. **AI explainability:** Analysts and facility managers must understand why every alert is flagged however deep learning models (Autoencoders, LSTMs) are often black box system. Finding a way to provide interpretable explanations of found anomalies in time-series and traffic data in an accurately efficient way is still an open issue. We hope to mitigate it with XAI and the potential for natural-language explanations.
4. **Standardization of Protocols in Datasets:** Standardization across IoT communication and security protocols remains limited. Although SentinaAI supports several common protocols (MQTT, CoAP, Modbus), variations in authentication methods, encryption strength, and data formatting introduce challenges during mixed-protocol integration. A fully standardized and secure communication model for IoT devices is still an open research area among manufacturers. Until such standards emerge, cross-protocol parsing and normalization will continue to add latency and may introduce additional security vulnerabilities.